

PSYCHOLOGY

Languages Pathway



**SENIOR
4**

STUDENT'S BOOK



Psychology

Senior



FOR

LANGUAGES PATHWAY

STUDENT'S BOOK

EXPERIMENTAL VERSION

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FOREWORD

Dear Student,

The Rwanda Basic Education Board is honoured to present to you this Psychology student book for Senior Four. It serves as a guide to competence-based teaching and learning, ensuring consistency and coherence in the study of human behaviour and mental processes. The Rwandan educational philosophy is to ensure that you achieve your full potential at every level of education, preparing you to be well-integrated into society and to exploit diverse employment opportunities.

The Government of Rwanda emphasises the importance of aligning teaching and learning materials with the syllabus to facilitate your learning process. Many factors influence what you learn, how well you learn, and the competences you acquire. These factors include the quality of instructional materials available and effective assessment strategies. Special attention has been paid to activities that facilitate the learning process, helping you develop your ideas and make discoveries through concrete psychological observations and experiments carried out individually or with your peers.

In a competence-based curriculum, learning is considered to be an active process of building knowledge and meaningful skills, values, and attitudes through experience and interaction with the environment. In this book, psychological concepts are mainly introduced through activities, real-life situations, or scenarios that help you construct knowledge, develop critical thinking skills, and acquire positive attitudes and values.

For effective use of this textbook, your role is to:

- Work on given activities which lead to the development of practical and analytical skills.
- Share relevant information with other students through presentations, discussions, group work, and active learning techniques such as role play, case studies, self-reflection, and research in the library, on the internet, or within your community.
- Participate and take responsibility for your own learning and personal growth.
- Draw conclusions based on findings from learning activities and psychological theories.

To facilitate your progress, the content of this book is self-explanatory, allowing you to use it independently to acquire and assess your competences. The book is organised into units, where each unit follows a logical learning path: it starts with a key unit competence, followed by an introductory activity to stimulate your interest. This leads into the development of psychological concepts connected to real-world situations, supported by various learning activities. Finally, each unit concludes with an end-of-unit assessment to help you evaluate your own progress and mastery of the subject matter.

I wish to sincerely extend my appreciation to the REB staff who organised the editing process of this textbook. Special gratitude also goes to the lecturers, teachers, illustrators, designers, and experts from both governmental and non-governmental sectors who supported the exercise throughout. Any comments or contributions would be welcome to improve the next edition of this textbook.



Dr. MBARUSHIMANA Nelson

Director General, REB

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I would like to extend my sincere appreciation to everyone who contributed to the development and refinement of this Psychology textbook for Senior Four. This achievement would not have been possible without the active involvement and expertise of various stakeholders committed to the advancement of social sciences in Rwanda.

Special thanks go to the Rwanda Basic Education Board (REB) staff, subject experts, university lecturers, and secondary school teachers for their valuable work in drafting and reviewing the pedagogical content of this book. Their dedication ensures that the concepts of human behaviour and mental processes are presented accurately and effectively for Rwandan students.

I am also grateful to all our education development partners whose technical and financial support was instrumental throughout the curriculum alignment and production process.



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Head of Curriculum, Teaching and Learning Resources Department/REB

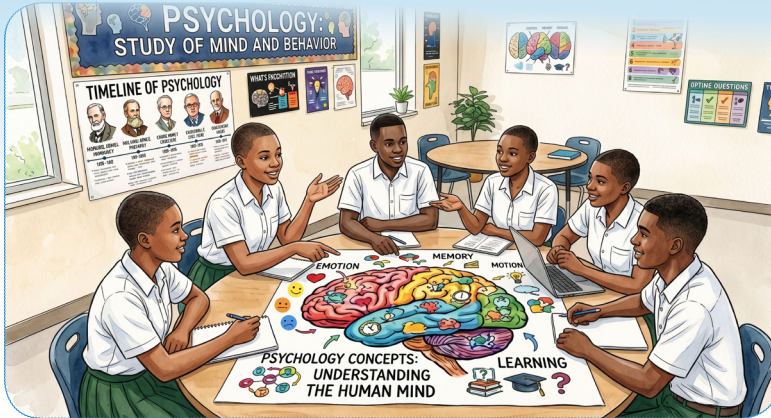
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UNIT 1

INTRODUCTION TO PSYCHOLOGY



Key Unit Competence:

Demonstrate an understanding of key psychological concepts, major figures, and the role of Psychology in life.

Learning objectives

By the end of this unit, you should be able to:

Knowledge and understanding

- Define key concepts of Psychology
- Describe the history of Psychology briefly
- Explain why Psychology is considered a science.
- Describe the contributions of major figures in Psychology
- Explain the goals and importance of Psychology

Skills

- Construct your own meaning of Psychology.
- Apply basic psychological concepts to explain everyday human behaviour.
- Use Psychology in our daily life.
- Defend the importance of Psychology in daily life

Attitudes and values

- Appreciate the relevance of Psychology in understanding themselves and others.
- Defend the relevance of Psychology in our daily life.
- Develop curiosity and an open mind toward scientific study of behaviour and mental processes.

Introduction

Have you ever wondered why people react differently to the same situation or remember the same events in different ways? Why do you sometimes respond quickly to a comment and only later ask yourself, "Why did I react that way?" Everyday experiences such as feeling nervous before speaking or presenting in class, suddenly forgetting something important, or becoming highly motivated to complete a task show that human behaviour is influenced by many hidden mental processes. These everyday moments show that there is more to human behaviour than what we can see on the surface.

Psychology helps us explore and explain these everyday experiences. It is the scientific study of human behaviour and mental processes, including how people think, feel, remember, learn, and respond to the world around them. In this unit, you will discover what Psychology is, how it has been developed as a scientific field, and how the ideas of important psychologists contributed to our understanding of human behaviour.

Through the study of key concepts such as behaviour, memory, emotion, motivation, and perception, you will learn how Psychology helps us understand ourselves and others. You will also see how psychological knowledge can be used to interpret everyday situations and make better decisions. By the end of the unit, you will be able to explain basic psychological concepts and relate them to real-life experiences.

Introductory activity



Exam moments case study: Candidates Umutoni and Rangira

Read the case study and answer the questions that follow.

On the morning of the S3 National Examination, Umutoni and Rangira stood outside the exam room. As the time approached, Rangira became visibly uneasy; his hands trembled, and he avoided eye contact and speaking to anyone. When Umutoni greeted him, he quietly said that he felt like he had forgotten everything he studied. Noticing his distress, Umutoni decided not to overwhelm him with more questions and stayed nearby without speaking.

After the exam, Rangira returned smiling and relaxed. He told Umutoni that once he started writing, everything he had studied came back to him, and the exam was easier than he expected.

1. What do you think was happening to Rangira before the exam, based on his behaviour and actions?
2. Why do you think Umutoni decided not to ask more questions and instead gave Rangira quiet support?
3. Which psychological ideas (such as emotion, behaviour, thinking, or memory) can help explain Rangira's reaction?
4. Have you ever felt stressed or overwhelmed before an exam or important activity? What similarities do you notice with Rangira's situation?
5. How can studying Psychology help you understand situations like this and improve your relationships with others?

1.1. Key Concepts of Psychology

Learning activity 1.1



Read the case study and answer the questions

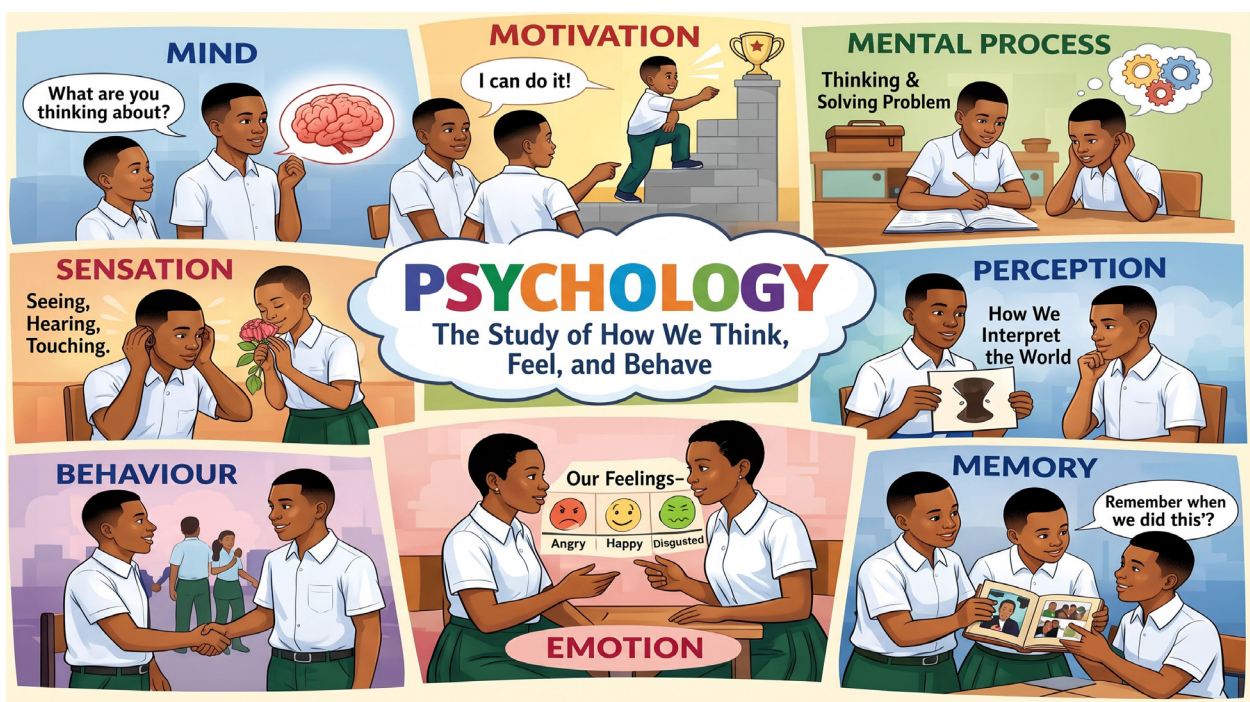
During sports activities in the school playground, Senior Four students were playing a football match with Senior Five students. When the ball moved toward a Senior Four player named Ntwari, Ntwari quickly noticed it and ran to kick it while hearing teammates calling out the name, Ntwari! Ntwari!, Ntwari!!. As Ntwari controlled the ball, different thoughts ran through Ntwari's mind: how to pass the ball correctly, how their team could score, and how winning would feel. Feeling motivated, Ntwari ran faster and tried his best to help the team.

While Ntwari was chasing the ball, He accidentally bumped into an opponent. The bumped player reacted quickly and looked upset, but Ntwari helped him to get up and apologised to him, which calmed the situation. Later, when Ntwari's team scored a goal, Ntwari celebrated with energy and excitement. After the match, Ntwari explained to classmates the strategies their team used to play well.

Questions

1. Based on Ntwari's actions, thoughts, and feelings during the game, what do you understand by the word "Psychology?"
2. How did Ntwari's thoughts influence Ntwari's behaviour during the game? Give one example.
3. What emotions are shown by Ntwari and the bumped opponent? How did these emotions shape behaviour?
4. How did social factors (teammates' shouts, opponent's response, class discussion) influence what happened?
5. Think about a similar moment from your life in games or class. Which psychological ideas help explain it?

Key concepts of Psychology are the fundamental ideas or main principles that explain how humans think, feel, and behave. They include concepts like mind, behaviour, emotion, memory, perception, sensation, motivation, and mental processes, which help psychologists understand and study human thoughts, feelings, and actions.



1.1.1. Psychology, mind and behaviour

The word **Psychology** is derived from two Greek words “psyche” and “logos.” Psyche means soul or spirit, and logos means the study. Early thinkers described Psychology as the study of the “soul” Modern Psychology developed into a scientific field. Today, Psychology is understood as the scientific study of behaviour and mental processes. This means that psychologists try to understand what people do, what goes on in the mind, and why individuals behave in certain ways. By observing actions and examining thoughts, emotions, memories, motivations, and decisionmaking, Psychology helps explain every day’s situations at school, at home, and in the wider community.

The mind involves internal processes that guide how a person thinks, feels, remembers, plans, solves problems, and makes decisions. These processes cannot be seen directly, but their effects appear in what people do. For example: before a debate, a student silently reviews key points, recalls facts, and plans how to speak. These mental processes thinking, memory, and planning are happening in the mind, even though others cannot see them.

Behaviour is any action or reaction that others can observe or measure, such as speaking, helping, moving away from danger, or physical responses like shaking or sweating when a person is nervous. Behaviour is influenced by emotions, culture, attitudes, values, peer influence, past experiences, and biological make-up. For example: during Community work (Umuganda), a student notices an elderly neighbour struggling to cut down bush and walks over to help. This visible action of helping is behaviour, which may be guided by values, empathy, and community expectations.

The mind influences behaviour; for example, fear may cause someone to avoid a situation (phobia), while behaviour provides clues about what the mind is processing, such as avoiding eye contact hinting at nervousness. By understanding this connection between internal mental processes and outward actions, Psychology helps explain why people act, think, and feel the way they do.

1.1.2. Mental processes, emotion and motivation

Mental processes are internal activities of the mind that enable a person to think, understand, learn, remember, imagine, reason, solve problems, and make decisions. Although they cannot be observed directly, they influence how individuals interpret information and react to different situations. Examples include thinking, memory, perception, and emotions, which help people make sense of experiences and guide their behaviour.

These processes operate continuously and shape many activities in everyday life. For instance, during community work, a resident may plan tasks and recall instructions from the local leader, while a market vendor compares prices and calculates change. The study of these mental activities helps psychologists seek to explain why people behave in certain ways and how this knowledge can support better decision-making, stress management, and responses to future situations.

Emotion refers to the feelings a person experiences in response to events or situations, such as happiness, fear, anger, or sadness. It involves a combination of awareness, bodily reactions, and observable behaviour. Emotions help people interpret and respond to the world by signalling what may be safe or dangerous, and what may bring happiness, sadness, or anger. In this way, emotions guide how individuals react and interact with others. For example: Before speaking in class, a student may feel nervous, which may cause a faster heartbeat, shaking hands, or avoiding eye contact, showing how emotions influence behaviour.

Motivation is the inner drive/acquired or reason that encourages a person to start, continue, or stop a behaviour in order to achieve a goal. It guides goal-oriented actions and helps individuals remain committed to what they want to accomplish. Motivation may come from personal goals, encouragement from others, or the desire to succeed. For instance, a student who wants to improve marks may study longer hours or seek help from a teacher.

Together, mental processes, emotions, and motivation help explain why people think, feel, and behave the way they do, making them essential concepts in Psychology and important for understanding everyday behaviour.

1.1.3. Memory, perception, and sensation

Memory is the process of acquiring, storing, and retrieving information. It involves three main stages: encoding, storage, and retrieval. Human memory allows people to retain and recall experiences or learned information. For example, remembering a lesson learned in class helps a student answer a question later.

Perception is the process through which the brain selects, organises, and interprets information received from the senses. People may perceive the same situation differently because the brain interprets sensory information based on past learning, memory, emotions, and expectations. Perception involves both bottomup processing, where understanding starts with the sensory details themselves; for example, when a child sees a new object like a toy's shape, colour, and size before recognising it. It also involves topdown processing, where interpretation is guided by what a person already knows, thinks, or expects from experience (e.g., recognising a familiar object in dim light).

Sensation refers to the inputs about the physical world obtained by our sensory receptors or received through our sensory organs (eyes, ears, nose, tongue and skin). It is the process by which an organism's sense organs respond to a stimulus and send nerve impulses to the brain. After reaching the brain, they are registered as a touch, a sound, a taste, and a splash of colour. Hence, sensation can be thought of as an organism's first encounter with sensory stimuli.

NOTE: Sensation is difficultly defined separately from perception. Sensation is the door to the physical world available to our sensory receptors, and perception is the process by which the brain selects, organises and interprets these sensations. In other words, sensation is the psychological basis of perception.

Example: Hearing sounds and seeing colours are sensory processes; whereas, listening to sweet music and detecting depth in a two-dimensional picture are perceptual processes. Without some kind of sensation, no perception could take place.

To understand more about sensation and perception, you can visit this website: <https://courses.lumenlearning.com/suny-hvcc-Psychology-1/chapter/outcome-sensation-and-perception/>

Application activity 1.1



Questions

1. What does Psychology study? Give one example showing the difference between behaviour you can see and a mental process you cannot see.
2. Choose one everyday life situation (Community work, market buying, helping at home). Explain it using one psychological idea (emotion, memory, motivation, perception).
3. How can learning Psychology help you be more patient and fairer with others in your school or community?
4. Match the following concepts with their meaning

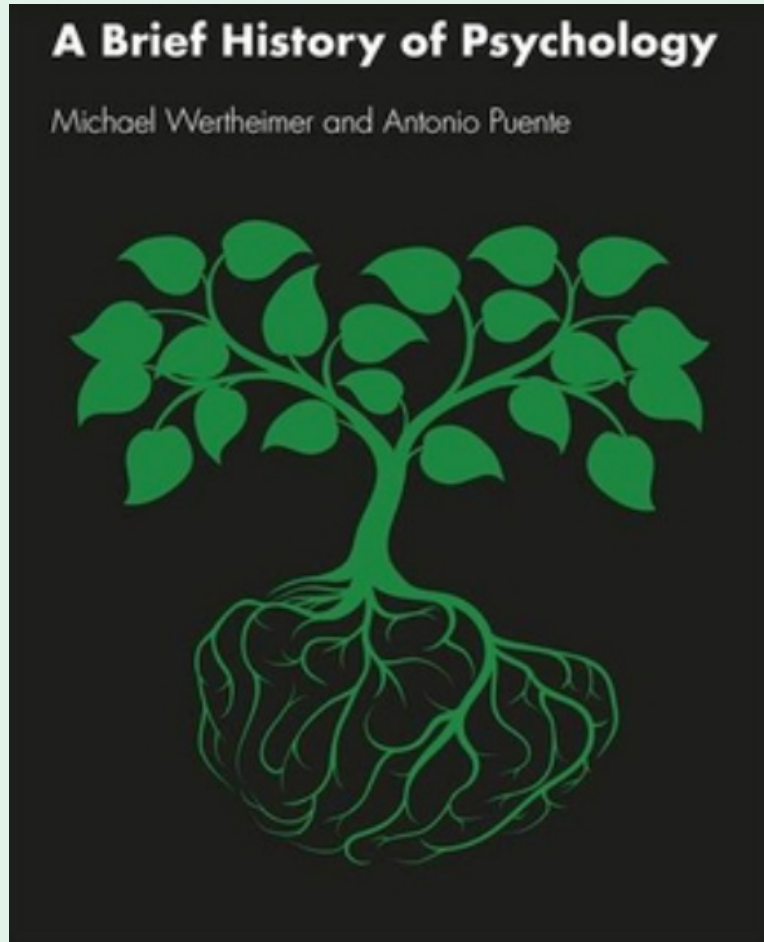
Concept	Meaning
1. Behaviour	a) Receiving information through our sensory organs
2. Mind	b) The process of acquiring, storing, retaining, and retrieving information.
3. Emotion	c) Process by which the brain selects, organises, and interprets information from our senses.
4. Memory	d) Actions or reactions of a person that can be seen or observed by others.
5. Motivation	e) Set of mental abilities that allow a person or animal to think, feel, remember and understand things
6. Perception	f) The force that drives a person to act, work or achieve a goal
7. Sensation	g) Feelings that a person experiences because of a situation or event.
8. Mental Process	h) The scientific study of behaviour and mental processes.
9. Psychology	i) Activities that happen in the mind which help an individual to think, understand, learn and remember

1.2. Brief history of Psychology

Learning activity 1.2



Exploring the Origins and History of Psychology



Source: <https://www.routledge.com/A-Brief-History-of-Psychology/Wertheimer-Puente/p/book/9781138284746>

Study the image above showing a tree with brain-shaped roots. Then think and answer the questions below.

1. What message do you think the tree with brain-roots image is trying to show about how Psychology began and how it has grown as a field?
2. In your own words, what do the words 'origin' and 'history' mean?
3. Why is it important for you to know the history of Psychology when starting to study the subject?

During an evening study session, your teacher places three objects on a table: a traditional drum, a notebook, and a photo of a person thinking deeply. Your teacher then asks: "Before Psychology became a science, who do you think observed human behaviour and tried to explain the mind?"

Psychology as a study of mind and behaviour has roots in ancient civilisations and has developed over many centuries. Early explanations of human behaviour were mainly

philosophical and often linked to cultural beliefs or supernatural ideas. With time, Psychology evolved into a scientific discipline that uses observation, measurement, and experiments to understand mental processes and behaviour. The development of psychological thought can be organised into four main parts:

1.2.1. Ancient Psychology and Scientific Psychology

Ancient Psychology

During this period, people used reasoning to understand the mind. The roots of Psychology can be traced back to ancient Greece, where philosophers such as Plato (427-347 BCE) and Aristotle (384-322 BCE) laid the groundwork for Western thought about the mind and behaviour.

Plato, a student of Socrates, believed in the existence of an eternal soul and proposed that knowledge is innate, meaning that humans are born with knowledge that unfolds over time. In contrast, Aristotle emphasised empirical observation and argued that knowledge comes from experience. His book *De Anima* (On the Soul) is considered one of the first systematic works on Psychology.

René Descartes (1596-1650) also proposed “mind - body dualism”, saying the mind controls the body. His idea became influential in the 17th century around 1641.

Hippocrates, the father of modern medicine, also made significant contributions to early psychological thoughts. He proposed the theory of the four temperaments, which linked personality traits to bodily fluids or “humours.” According to Hippocrates, an imbalance of blood, yellow bile, black bile, or phlegm could lead to psychological disturbances.

Together, these ideas prepared the ground for Psychology’s scientific turn in the 19th century, when later scholars would adopt measurement and experimentation to study the mind and behaviour.

Scientific Psychology

During the period of 1879, Psychology became a science when Wilhelm Wundt opened the first Psychology laboratory in Germany. It was then that Psychology started using experiments and scientific methods to study the mind.

School of thought (19th-20th century)

Psychology has evolved through several influential schools of thought, including structuralism, behaviourism, cognitive Psychology, psychoanalysis, humanism, Gestalt, and functionalism. Each one offers unique insights into the mind and behaviour. Today’s psychologists often draw from multiple schools of Psychology rather than aligning strictly with one.

Structuralism

Edward B. Titchener believed that Psychology should study the human mind. He believed that the mind can be broken into basic elements such as sensations, feelings, and images. He used a method called introspection, where people describe their own mental experiences; for example, a student who just failed an exam sits quietly and thinks: “Why did I fail? Did I not study enough, was I nervous during exams?” By reflecting on his/her own thoughts, feelings, and actions to understand the cause of that failure, that student is practising introspection.

Any one of several schools of 20th-century linguistics committed to the structuralism principle that language is a self-contained relational structure, the elements of which derive their existence and their value from their distribution and oppositions in texts or discourse. This principle was first stated clearly for linguistics by the Swiss scholar Ferdinand de Saussure (1857–1913). Ferdinand de Saussure developed structuralism by explaining that language is not just a simple collection of words, but a structured system in which all elements are connected and depend on one another. He argued that meaning does not come from individual words alone, but from the relationships and differences between them, such as how “cat” gains meaning by being different from words like “bat” or “cap.” He also introduced the idea that language has an underlying structure made up of rules, which he called *langue*, while the actual spoken or written use of language is called *parole*, with structuralism focusing mainly on *langue*.

In addition, Saussure explained that linguistic signs are made up of a signifier (the sound or written form) and a signified (the concept in the mind), and that meaning is created through the relationship between these two parts within the overall system. Together, these ideas show why he viewed language as a structured network where meaning is relational rather than independent

Functionalism in 1890

William James (January 11, 1842 – August 26, 1910) was an American philosopher, historian, and psychologist, and the first educator to offer a Psychology course in the United States. He developed functionalism in 1890 and emphasised how the mind helps people to adapt. He believed that Psychology should study how the mind functions and how it helps people to adapt to their environment. He focused on the purpose of behaviour and mental processes and emphasised practical applications of Psychology in everyday life.

Psychoanalysis

Sigmund Freud (1856–1939) is considered the founder of the psychodynamic approach to Psychology, which seeks to explain human behaviour through unconscious drives. Freud believed that the mind is responsible for both conscious and unconscious decisions that are made based on psychological drives (the unconscious mind influences human behaviour, and that past experiences, especially childhood experiences, affect behaviour). Sigmund Freud explained personality using 3 parts: the **id** (basic desire), **ego** (rational thinking), and **super-ego** (moral value) to comprise a person's personality. Freud believed that people are “simply actors in the drama of their own minds, pushed by desire, pulled by coincidence. Underneath the surface, our personalities represent the power struggle going on deep within us; for example, a person may feel fearful or anxious because of hidden memories or experiences.

Behaviourism:

John Broadus Watson (1878–1958) was an American psychologist who founded behaviourism, the idea that Psychology should focus on observable actions rather than thoughts or feelings. He showed that emotions like fear could be learned through conditioning, most famously in the Little Albert experiment, and his work shaped Psychology, education, and advertising. He believed that behaviour is learned through interactions with the environment and can be shaped or controlled using conditioning techniques.

Humanism:

It is a school of Psychology developed mainly by Carl Rogers in the 1950s and Abraham Maslow in 1943–1954. Humanistic psychologists believe that human beings have free will and can make their own choices in life. They also believed that people are naturally good and have a strong desire for personal growth and self-development. According to Maslow, every person is motivated to achieve self-actualisation, which means becoming the best version of themselves and fully developing their talents and abilities. Humanism therefore focuses on understanding human potential, personal experiences, and the importance of helping individuals grow and achieve their goals.

Gestalt Psychology

Gestalt Psychology is a school of Psychology that focuses on how people perceive and understand whole patterns rather than separate parts. It was developed by Max Wertheimer in 1912, together with Wolfgang Köhler and Kurt Koffka. Gestalt psychologists believed that the mind organises sensory information into meaningful wholes, and that the whole experience is different from simply adding its parts. They emphasised that human perception naturally groups objects, shapes, and events in ways that make them easier to understand. This approach helped explain how people perceive patterns, solve problems, and understand their environment.

For example, through:

- a) **The law of closure**, we can recognise a complete shape even when it is not fully drawn, such as seeing a full circle even if parts are missing.
- b) **The law of proximity** shows that objects that are close to each other are perceived as a group, while the law of similarity explains that items that look alike are grouped by the mind.
- c) **The law of figure-ground** allows us to distinguish an object from its background; when driving or walking in town, you can easily separate a car from the background of buildings and trees
- d) **The law of continuity** shows that we tend to perceive smooth and continuous patterns rather than broken or disjointed lines.

1.2.2. Modern Psychology (1950-present)

Modern Psychology is the scientific study of the human mind and behaviour using empirical research, advanced tools, and interdisciplinary approaches. It focuses on understanding how people think, feel, learn, and behave in various environments, including social, cultural, and digital settings.

In other words, modern Psychology uses scientific research, experiments, observation, and application in real life. The American psychologist and social scientist Burrhus Frederic Skinner developed the reinforcement theory (1940s-1950s) and was one of the most influential psychologists of the 20th century.

Skinner pioneered the science of behaviourism, discovered the power of positive reinforcement in education, invented the Skinner Box, and designed the foremost psychological experiments that gave predictable and quantitatively repeatable outcomes. Jean Piaget, in the 1930s- 1970s, developed cognitive development theory, and Carl Rogers developed humanistic Psychology in the 1950s

Application activity 1.2



1. Explain what a school of thought in Psychology is, and give two examples of Schools of Psychology.
2. Study the situation/ description from the table below and identify which stage of the history of Psychology it represents.

Description/situation	Part of history of Psychology	Explanation
1. Early thinkers used reasoning and Philosophy to explain the mind and behaviour. Philosophers like Plato and Aristotle discussed how the mind works	I. Modern Psychology	A. Psychology became a scientific discipline when Wilhelm Wundt started
2. In 1879, a psychologist opened the first laboratory to study the mind using experiments.	II. Scientific Psychology	B. These psychologists created major psychological approaches.
3. Psychologists began developing different approaches to explain behaviour and mental processes, such as psychoanalysis and behaviourism.	III. Ancient (philosophical) Psychology	C. Different psychologists developed theories explaining how the mind and behaviour work.
4. Psychologists study how people think, remember, and solve problems using scientific research and technology.	IV. Schools of thought.	D. Psychology was part of Philosophy and relied mainly on reasoning and ideas.
5. Different psychologists such as Sigmund Freud and John B. Watson developed theories to explain human behaviour.	V. Schools of thought and experimental research.	E. Modern Psychology uses scientific methods and research to study behaviour and mental processes.

1.3. Scope of Psychology

Learning activity 1.3



1. Read the situation below and answer the question

"A teacher asked students to memorise new vocabulary words. Some students still remembered the words quickly while others took more time to remember those words. During the group discussion, they expressed ideas and exchanged their opinions". Which mental abilities were used when students were memorising vocabulary words?

2. As a student, explain when you applied Psychology in decision making.

The scope of Psychology refers to the range of areas that Psychology studies. Psychology focuses on how people think, feel, learn, remember and behave in different situations. Psychologists such as Wilhelm Wundt helped establish Psychology as a scientific field that studies these aspects of human life. In linguistics, it shows how Psychology helps us to understand language learning, memory, thinking, reading, writing, and communication.

1.3.1. Main Areas of Study of Psychology

Study of human behaviour

Psychology studies how people act in different situations, including their habits, reactions, and social interactions. For example, a person might help a stranger in need, showing kindness, while another may avoid helping due to fear or self-interest.

Study of mental processes

This scope focuses on how people think, remember, perceive, and solve problems. For example, a student solving a difficult math problem or remembering instructions from a teacher.

Study of growth and development

Psychology studies how people grow and change from childhood to adulthood. It examines physical, emotional, social, and intellectual development over time. For example, a baby learning to walk, a child learning to read, or a teenager developing reasoning skills are all part of human development.

Study of individual differences

Psychology studies how people differ from one another. Some people learn quickly while others learn slowly. Some people are more emotional or more social than others, or someone may be an introvert while another one is an extrovert.

Emotions and Motivation: Psychology explores feelings, drives, and motives that influence behaviour. For example, winning a sports match may make a person feel happy and motivated to train harder, while fear of failure might cause hesitation.

Social Interactions: This scope studies how people relate to each other in groups and

society. For example, peer pressure can influence a teenager to join a club, or group cooperation can help achieve a common goal.

Learning and Education: Psychology helps understand how people acquire knowledge and skills. For instance, using repetition, flashcards, or interactive teaching helps students remember information better and learn new skills efficiently.

Applied Psychology: This area focuses on using psychological knowledge in real-life settings such as health, workplaces, law, and sports. For example, a clinical psychologist helps patients with depression, or a sports psychologist helps athletes improve performance.

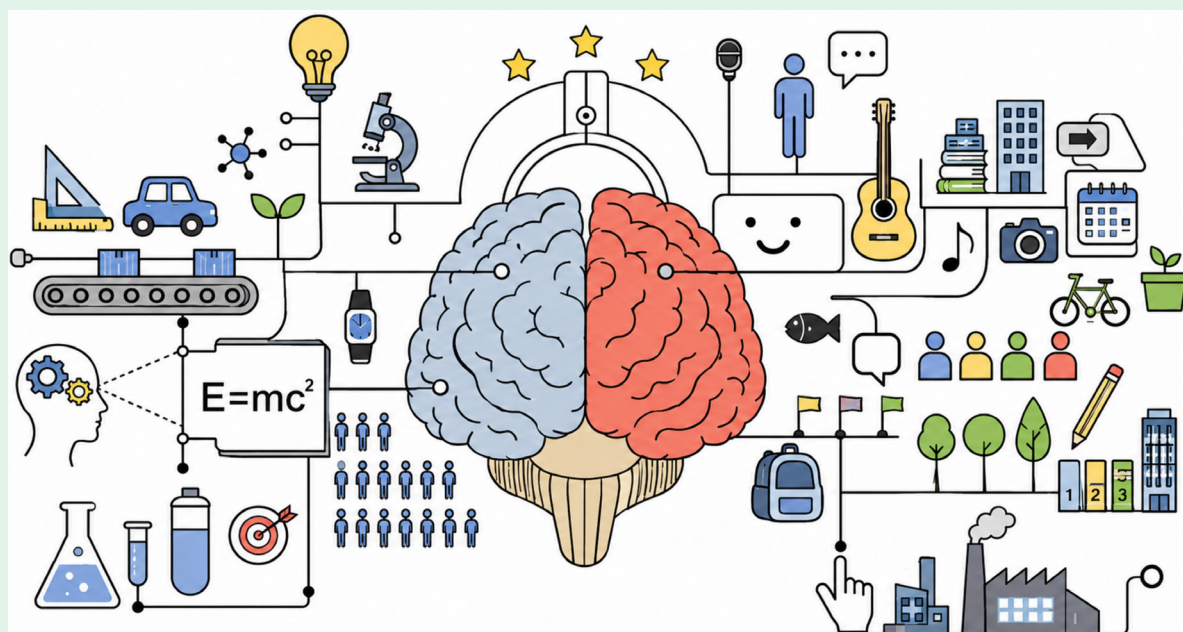
Application activity 1.3



1. Illustrate the difference between Psychology and the scopes of Psychology based on meaning and focus, and give an example.
2. Using an example, explain which scopes of Psychology affect your learning more as a language student.

1.4. Psychology as a Science

Learning activity 1.4



Source: <https://studiousguy.com/Psychology-and-other-disciplines/>

1. Observe the above picture and make a summary note of what you see.
2. What do you understand by the word "Science" and how is it related to Psychology?

Psychology is the scientific study of the mind and behaviour in humans and animals. It examines both conscious and unconscious processes, including thoughts, feelings, and motivations. The field of Psychology is broad and connects the natural sciences with the social sciences.

Biological psychologists study how the brain and nervous system influence behaviour, linking Psychology with neuroscience.

Social psychologists focus on understanding how individuals behave within groups and social environments.

Psychologists also explore mental processes such as perception, attention, emotion, intelligence, and personality. Their work helps explain how people think, feel, and act in different situations. Overall, Psychology seeks to understand the relationship between mental functions and behaviour.

Several psychologists helped establish Psychology as a science by using experiments, observation, and systematic research methods to study behaviour and mental processes. Important ones include:

Wilhelm Wundt believed that Psychology should be studied scientifically through experiments and observation. In 1879, he established the first Psychology laboratory in Germany, where he conducted experiments to study mental processes such as sensation, perception, and reaction time. He introduced the method of introspection, where individuals described their conscious experiences. His work marked the beginning of Psychology as an independent scientific discipline. It was in this year that Psychology became a science.

William James supported the idea that Psychology should study the functions of the mind in a scientific way. He believed that the purpose of mental processes is to help individuals adapt to their environment. Through his approach known as functionalism, he emphasised studying how thinking, learning, and emotions help people survive and adjust to everyday life.

John B. Watson argued that Psychology should be a purely objective science that focuses only on observable behaviour rather than internal mental processes. In 1913, he introduced behaviourism and insisted that psychologists should study behaviour through experiments and measurable observations. He believed that behaviour is learned through interaction with the environment.

B. F. Skinner further developed the scientific study of behaviour by conducting controlled experiments on learning. He believed that behaviour is shaped by reinforcement and punishment, and that scientific experiments can be used to predict and control behaviour. His research strengthened the idea that psychologists can study behaviour using scientific methods.

Application activity 1.4



1. What do you understand by the term "Psychology as science"
2. Using thoughts from different psychologists, explain why Psychology is considered a science.
3. In which ways does Psychology as a science connect with other subjects you study at school?

1.5. Main figures in Psychology

Learning activity 1.5

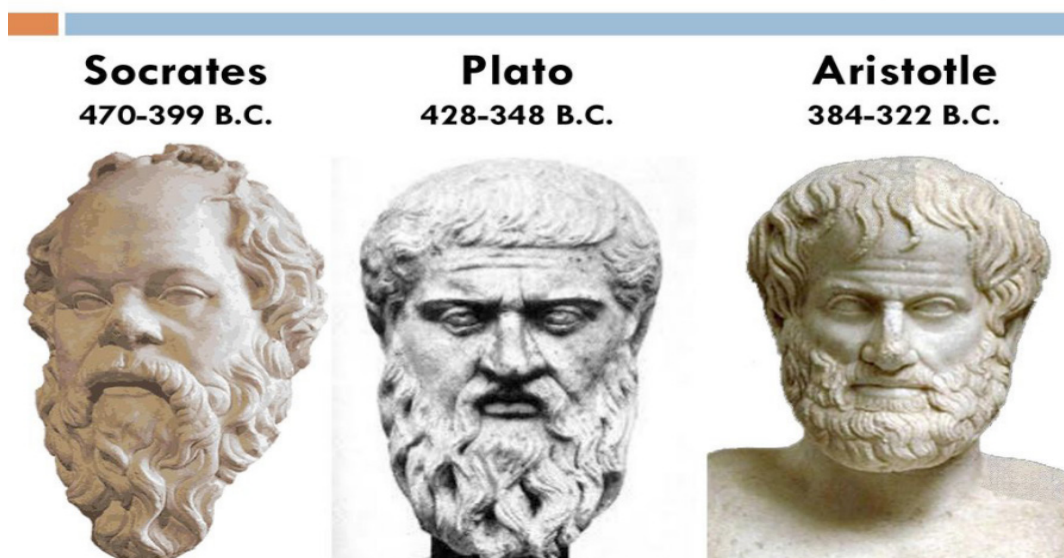


1. Can you recall the philosophers who introduced Psychology?
2. Who is claimed to be the first Psychologist?

In Psychology, the term “main figures” refers to the most influential psychologists or thinkers who have made foundational contributions to the study of the mind, behaviour, and mental processes. These individuals are recognised for introducing key theories, methods, or discoveries that shaped the field of Psychology.

1.5.1. Socrates, Plato and Aristotle (Greek philosophers)

Greek Philosophers



Source: <https://slideplayer.com/slide/13030639/>

The ancient Greeks partially paved the way for the scientific study of Psychology and the understanding of certain psychological problems. In fact, Aristotle can be credited with the formulation of the foundations of Psychology. Since Greek philosophers studied how the personality and character of humans expressed themselves either as part of rational and deductive processes or as altered irrational processes, it should come as no surprise that Aristotle combined Psychology with philosophy of mind, and therefore his empirical approach was a precursor to the modern psychological approach.

1.5.2. Wilhelm Wundt (1832-1920) and Sigmund Freud (1856-1939)



Father of Psychology

Started the first laboratory to study humans in Leipzig, Germany 1879.

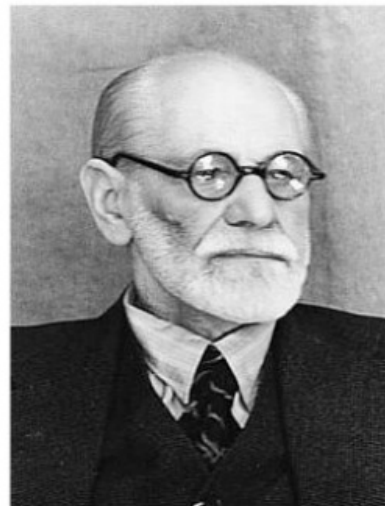
Tried to apply scientific method to the study of the mind.

Source: <https://slideplayer.com/slide/10171045/>

Wilhelm Wundt is widely credited with establishing Psychology as an independent scientific discipline. In 1874, he wrote the first major Psychology textbook, *Principles of Physiological Psychology*. In 1879, he founded the first laboratory dedicated to psychological research at the University of Leipzig. Before this, Psychology was mainly studied under philosophy and biology. Wundt also introduced the experimental use of introspection (self-examination), which is why he is often called the father of experimental Psychology, and many scholars consider him the founder of modern Psychology.

Sigmund Freud - History

- May 6, 1856-Sept 23, 1939
- "Father of Psychoanalytics"
- University of Vienna
- was an Austrian neurologist and psychiatrist who co-founded the psychoanalytic school of psychology.
- known for his theories of unconscious mind and redefinition of sexual desire as mobile and directed towards a wide variety of objects.



Source: <https://www.pinterest.com/pin/749356825469971381/>

Sigmund Freud is one of the most famous figures in Psychology and the founder of psychoanalysis. He studied the unconscious mind and developed talk therapy. Freud proposed that the mind consists of the id, ego, and superego, and introduced ideas such as defence mechanisms and psycho-sexual stages of development. Although later approaches reduced his influence, his work laid the foundation for modern psychotherapy.

1.5.3. Mary Whiton Calkins (1863 - 1930) and Jean Piaget (1896-1980)



- First female to complete all PhD requirements at Harvard
- Was denied degree because of gender
- First female president of the American Psychological Association

Source: <https://slideplayer.com/slide/13017596/>

It is no secret that women make up the majority of psychologists. The women in the early days of Psychology paved the way for the women of today. Although there were several women pioneers, none was more fundamental than Mary Whiton Calkins. She was the second woman to complete the work necessary for a PhD in Psychology but was not awarded the degree because Harvard did not formally admit women at the time. She later became the first woman to serve as the president of the American Psychological Association. Besides being a trailblazer, her scientific contributions were robust. She invented the paired association technique of memory. In addition, she was one of the first to believe that the study of Psychology should be based on the conscious self, seen in relation to its environment.

Cognitive development theory

According to Jean Piaget,

- Children “construct” their understanding of the world through their active involvement and interactions.
- Studied his 3 children to focus not on what they knew but how they knew it.
- Describe children’s understanding as their “schemas” and how they use:

- **Assimilation:** Assimilation happens when a person uses an existing idea or knowledge to understand something new, without changing that idea.

For example, a child knows that a dog is a four-legged animal that barks. When the child sees a cow for the first time, they call it a “big dog” because it has four legs and makes the sound “moo”.

- **Accommodation:** Accommodation happens when a person changes or adjusts their existing idea or understanding after learning new information.

Jean Piaget



Source: <https://www.jf21shop.top/Product-Detail.aspx?iid=89421546&pr=52.88>

Example:

After being told that the animal is a cow, not a dog, and learning the differences (cow makes sound moo, dogs bark), the child changes their thinking.

Piaget developed the first theory of child cognitive development. At the time, it was groundbreaking. Before his work, children were believed to share the cognitive processes of adults. His theory directly led to the emergence of cognitive and developmental Psychology. He believed cognitive development was an active process, not something that children passively acquired. Piaget coined the popular psychological phrases schema, accommodation, and assimilation. In addition, he was a fierce child advocate and fought for the education of children as vital to a society's success.

Application activity 1.5



1. Who is the first female president of the American Psychological Association (APA)?

2. Study the table below and fill it appropriately.

Name of figure	Life date	Key contribution/theory	Real-life application
1. Wilhelm Wundt	I. 1896-1980	A. Cognitive development theory; concepts of schema, assimilation, accommodation; active learning	7. Emphasised observation and experience; precursor to modern methods
2. Socrates	II. 384 - 322 B.C	B. Empirical observation of mind and behaviour; studied rational and irrational processes; early Psychology foundations	8. Developed talk therapy; influenced modern psychotherapy techniques
3. Plato	III. 1856-1939	C. Psychoanalysis; unconscious mind; id, ego, superego; defense mechanisms; psychosexual stages	9. Influences teaching methods and early childhood education; emphasised how children learn differently from adults
4. Aristotle	IV. 428-348 B.C	D. Father of experimental Psychology; developed introspection; first Psychology lab	10. Wrote dialogues exploring philosophy; influenced Western thought and idealism
5. Sigmund Freud	V. 1832-1920	E. Theory of Forms: material world reflects higher perfect realities; ethics, justice, knowledge	11. Famous quote: "The unexamined life is not worth living"; inspired critical thinking and ethics
6. Jean Piaget	VI. 470-399 B.C	F. Socratic Method: questioning to uncover truth; virtue comes from knowledge; self-examination	12. His lab experiments laid the groundwork for modern research methods

1.6. Goals and importance of Psychology

Learning activity 1.6



1. What do you think psychologists are trying to achieve when they study human behaviour and mental processes?
2. Why do you think Psychology is among the subjects you learn?

1.6.1. Goals of Psychology

Psychology 1e :: by Charles G. Morris and Albert A. Maisto Chapter 1

GOALS OF PSYCHOLOGY

- Describe**
 - first goal of psychology is to describe the different ways that organisms behave
- Explain**
 - second goal of psychology is to explain the cause of behavior
- Predict**
 - third goal of psychology is to predict how organisms will behave in certain situations
- Control**
 - the fourth goal of psychology is to control an organism's behavior

MORRIS 1e © 2006 Prentice Hall

<https://slideplayer.com/slide/255327/>

Describe.

The first goal of Psychology is to describe behaviour and how people think. Psychologists do this by observing and measuring actions, like watching children play. By describing behaviour, we can understand what is normal and abnormal. For example, children talk, compete, use skills, and form groups while playing. They also notice and measure the abilities of others to work well together and succeed.

Explain

The second goal of Psychology is to explain why people behave the way they do. Psychologists ask questions like why someone acts a certain way or what affects personality, social behaviour, or mental health. They use theories, which are ideas that help explain the information they gather. For example, a friend might not answer a greeting for many reasons, like being busy, upset, or worried. Psychologists test their theories with the scientific method to understand and predict behaviour better.

Predict

The third goal of Psychology is to predict how people think and act by understanding their behaviour. When psychologists know why something happens, they can guess when it might happen again.

Modify

Modify behaviour and mental processes (Control)

The final goal is to apply this knowledge to improve people's lives and modify behaviour. For example, businesses use Psychology to design ads, and psychologists help people manage mental health problems. Psychology affects many parts of life, like education, work, leadership, and solving conflicts.

1.6.2. Importance of Psychology

The importance of Psychology daily



Image illustrating importance of Psychology daily

Psychology is the scientific study of human behaviour and mental processes that started long ago but became a separate science in the 1870s. It helps us understand why people act the way they do and how to deal with problems like stress and relationships. Psychologists work in many fields, not just therapy, including education, marketing, and justice. Psychology is important because it helps people make better decisions and manage life challenges in different ways, such as:

Building relationships: Psychology facilitates coexistence with others by understanding them better and working with their behaviour. By studying how people think, feel, and act, we can interact more effectively, reduce conflicts, and build positive relationships. Psychology gives us tools to understand why people behave the way they do, and how to respond in ways that promote harmony.

Improve communication: A better understanding of how humans think and behave will help people communicate better. They will be more effective in understanding what a person really means with their gestures and actions.

Build self-confidence: By better understanding yourself and your personality, you can gain more self-confidence. You will learn more about your weaknesses and be able to exploit them. Understanding your strengths, weaknesses, and personality traits helps you become more confident because you can make decisions and act in ways that suit you best. Understanding your weaknesses doesn't mean being negative; it means finding ways to manage them and turn them into advantages.

Enriching careers: You will be able to better understand your colleagues and have more chances to make friends. It helps to deal with your actions to enrich your career.

Importance of Psychology in a digital world



In today's digital world, where social media plays a big role, Psychology is very important for understanding how online interactions affect our minds and behaviour. Social media can influence emotions, self-esteem, and relationships, sometimes causing stress or anxiety.

Psychology helps us learn how to use social media in healthy ways, recognise harmful effects like addiction or cyberbullying, and develop better communication skills online. It also guides creators and businesses to design content that supports positive mental health. Overall, Psychology helps people navigate the challenges and opportunities of the digital age safely and wisely.

Application activity 1.6



1. Explain 3 goals of Psychology
2. How does Psychology help you to build self-confidence?

1.7. End unit assessment



Section A: Knowledge and understanding

1. Define Psychology.
2. Explain why Psychology is considered to be a science.
3. State two goals of Psychology.
4. The development of Psychology was influenced by early thinkers and the study of mental processes:
 - a) Identify two early philosophers who contributed to the development of Psychology.
 - b) Mention two mental processes studied in Psychology.
5. Read the situation below and answer the questions.

Alice failed her mathematics test. She became sad and refused to talk to her friends because some were mocking her. She decided to study harder for the next test.

- a) Identify one emotion shown by Alice. Identify one behaviour shown in the situation.

Section B: Skills

6. Explain how to apply, analyse, and use psychological concepts in real-life situations.

Section C: Attitudes and values

7. Explain one way in which Psychology can help you to improve a situation.
8. Through examples, explain four main goals of Psychology and show how it helps us understand the effects of social media on teenagers and youths

UNIT 2

SCHOOLS AND BRANCHES OF PSYCHOLOGY



Key Unit Competence:

Justify how key schools of Psychology support different branches of Psychology, and apply those branches to everyday life.

Learning objectives

By the end of this unit, you should be able to:

Knowledge and understanding

- Explain different thoughts in schools of Psychology
- Describe the key points of different schools of Psychology
- Explain different branches of Psychology
- Summarise different thoughts found in schools of Psychology
- State the implications of schools and branches of Psychology in society

Skills

- Compare and contrast different schools of Psychology based on their focus, methods, and assumptions.
- Classify examples of human behaviour according to the appropriate school or branch of Psychology.
- Make a reflection on others' works and ideas to sort out which is best for societal benefits.
- Analyse real-life cases using psychological perspectives to explain human behaviour and language learning

Attitudes and values

- Demonstrate respect for individual differences of thoughts, emotions, and behaviours
- Value the contributions of psychologists and psychological theories to understand learning and communication.
- Appreciate the knowledge of the major branches of Psychology
- Show openness to diverse perspectives and approaches within Psychology
- Recognise that human behaviour can be understood in multiple ways.

INTRODUCTION

Everyday situations such as feeling nervous before speaking in public, becoming motivated to complete a task, or misunderstanding someone during communication show that human behaviour is influenced by many mental processes. These experiences remind us that human thoughts, emotions, and actions can be explained in different ways. Psychology helps us understand these experiences. Over time, psychologists have developed different schools of thought to explain how people think, feel, and behave.

In this unit, you will explore major schools of Psychology such as structuralism, functionalism, psychoanalysis, behaviourism, and humanistic Psychology. Each of these schools provides a unique perspective on understanding human behaviour.

You will also learn about important branches of Psychology, including developmental, educational, cognitive, social, Family, clinical, and forensic Psychology. These branches focus on different aspects of human life and help explain behaviour in areas such as learning, relationships, communication, and personal development.

By studying these schools and branches of Psychology, you will understand different perspectives on human behaviour and language learning, respect individual differences, and apply psychological knowledge to everyday life.

Introductory activity



Think about times when you or your friend behaved in a way that was puzzling or interesting, then answer the following questions.

1. Why do people behave differently in different situations?
2. How do people learn new things or change their habits?
3. A bad situation has happened in the classroom suddenly; the school headteacher picks you randomly to explain what has just happened. How do you feel? Explain your answer.

2.1. Introduction to schools of Psychology: Structuralism, functionalism, psychoanalysis

Learning activity 2.1



1. Reflecting on the different ideas of psychologists you learned in Unit 1: Introduction to Psychology, what aspects of the mind might psychologists focus on when trying to understand a person's behaviour?
2. What do you know about these psychologists:
 - a) William James
 - b) Sigmund Freud
 - c) Wilhelm Wundt

Psychology developed over time as different psychologists tried to explain how the human mind works and why people behave the way they do. These different ways of thinking about the mind and behaviour are called schools of Psychology. A school of Psychology is a group of ideas or theories shared by psychologists who study the same aspects of human behaviour and mental processes. Each school has its own explanation of how the mind works. Among the earliest and most important schools of Psychology are Structuralism, Functionalism, and Psychoanalysis.

2.1.1. Structuralism

Edward Titchener (1867-1927)

- Titchener refined Wundt's technique of introspection and to study sensation and it **Structuralism**.
- He defined this as the study of the structure of the conscious mind.
- Titchener translated Wundt's major work "Principles of Physiological Psychology" into English.
- He considered himself a "true Wundtian" all his career.



Source: <http://slideplayer.com/slide/14138348/>

Structuralism is a school of Psychology that studies the "structure" of the mind by analysing conscious experiences using methods like introspection, where people describe exactly what they feel or think during a mental task.

Structuralism was the first school of Psychology, established in Germany by Wilhelm Wundt and later developed by his student Edward B. Titchener in England. It focused on breaking down the adult mind into its basic parts to understand how these parts combine to form complex thoughts and experiences. Wundt compared this process to how a chemist analyses chemical compounds.

The school aimed to identify the basic elements or "structures" of psychological experience, focusing on conscious experiences divided into sensations (sights, sounds, tastes, smells, and touch) and feelings (love, fear, joy, and sadness). The main method used was introspection, where participants described their experiences in detail while performing tasks, such as reading sentences, viewing colours, or solving math problems.

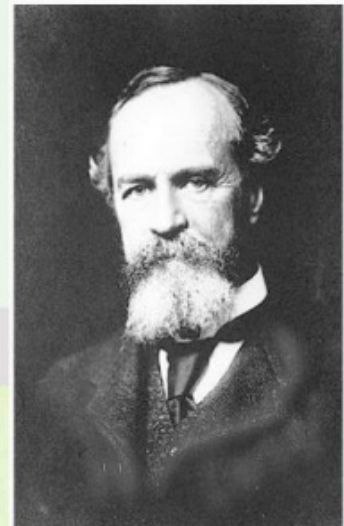
Structuralism helped Psychology become a science by showing that mental experiences could be studied systematically and emphasised the importance of conscious experience. However, it had limitations because participants often could not accurately describe their own experiences, and many mental events occur outside of awareness. Despite these limitations, structuralism laid the foundation for later schools of Psychology. An example of introspection is presenting a sentence on a card and asking a person to describe how it makes them feel, what they notice first, and what thoughts come into mind, mapping the elements of consciousness.

2.1.2. Functionalism

Functionalism is a school of Psychology founded by William James in the United States. It focuses on how the mind helps people and animals adapt to their environment rather than just studying thoughts. Functionalists were influenced by Charles Darwin's theory of evolution, which shows that traits helping survival are important. They also rejected introspection and studied how behaviour and mental processes are useful in everyday life.

William James (1842–1910)

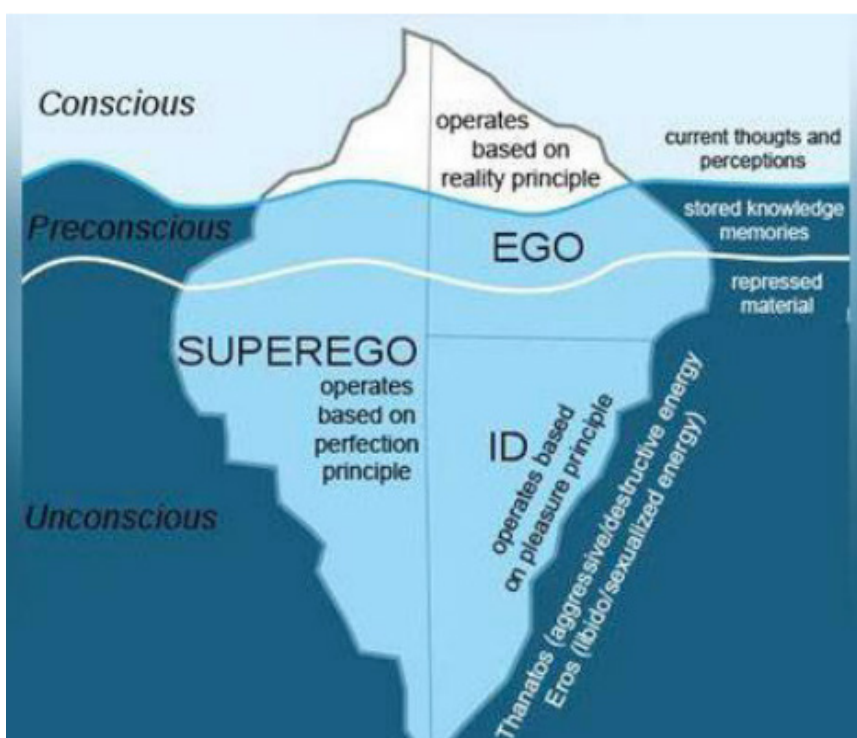
- First American psychologist
- Started psychology at Harvard in 1870s
- Opposed Wundt and Titchener's approach
- Author of the first psychology textbook
- Founder of Functionalism
 - functionalism – influenced by Darwin to focus on how behaviors help us adapt to the environment



Source: <https://www.slideserve.com/ashlyn/chapter-1-the-evolution-of-Psychology>

Functionalism also looked at how society works. It explained how family, education, religion, and media help people learn values and follow rules. Functionalists believed that shared values and habits keep society stable and organised. Even crime and deviance were studied as things created by society. Functionalism no longer exists as a school, but its ideas about adaptation, habits, and useful mental functions still influence Psychology today.

2.1.3. Psychoanalysis



Source: <https://www.indiatoday.in/education-today/gk-current-affairs/story/sigmund-freud-338418-2016-09-23>

Psychoanalysis is a school of Psychology founded by Sigmund Freud. It explains human behaviour by focusing on the unconscious mind, which contains thoughts, memories, and desires that people are not aware of. Freud believed that many human actions are influenced by hidden conflicts and past experiences. His theory is based on psychodynamics, which explains how mental energy is transformed into behaviour.

Freud explained that personality has three main parts: id, ego, and superego. The id seeks immediate pleasure and satisfaction of basic needs such as hunger and thirst. *For example*, if you are very hungry during class and see someone's snack on the desk, the id may push you to take and eat it immediately without thinking. The ego controls these desires by considering reality and what is acceptable in society. *For example*, the ego reminds you that taking someone's snack is wrong and could get you in trouble, so you decide to wait until break time arrives so that you can eat your own food.

The superego represents the moral values learned from parents and culture. *For example*, the superego makes you feel that stealing the snack would be morally wrong because you were taught to respect other people's property.

The interaction of these three parts influences a person's behaviour and personality.

Application activity 2.1



1. What do you understand by the term "school of Psychology"?
2. What is the difference between structuralism and functionalism?
3. In many Rwandan communities, people participate in Umuganda, where they work together to clean the environment and build community projects. During this activity, people may experience different sensations and feelings, and their behaviour helps them cooperate and adapt to community life.

Question:

Using your knowledge about functionalism, structuralism, and psychoanalysis, explain how people's sensations and feelings during community work can help us understand human behaviour.

2.2. Introduction to schools of Psychology: behaviourism and humanistic

Learning activity 2.2



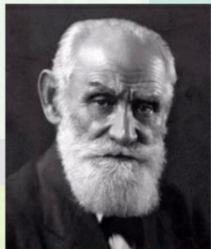
1. What do you remember from the following names?
2. John B. Watson, Ivan Pavlov, B. F. Skinner, Abraham Maslow, and Carl Rogers?

Over time, psychologists have been developing different Schools of educating people ways to understand human behaviour and mental processes, called Schools of Psychology. Each school uniquely looks at the mind and focuses on different ideas. Some schools include behaviourism, functionalism, and psychoanalytic Psychology.

2.2.1. Behaviourism

Ivan Pavlov (1849-1936)

- Behaviorist
- Russian Physiologist
- Studied learning through associations in animals
- Emphasized the study of observable behaviors



Behaviorism
John Watson & B.F. Skinner

- Watson focused on Classical Conditioning
 - Stimulus – stimulus learning
- Skinner is known for his work on Operant Conditioning
 - Response – Consequence learning



Watson (1878 -1958) Skinner (1904 -1990)

Source: <https://www.slideserve.com/ashlyn/chapter-1-the-evolution-of-Psychology>

Behaviourism is a school of Psychology developed by psychologists such as John B. Watson, Ivan Pavlov, and B. F. Skinner. It explains that behaviour is learned from the environment through experience.

Pavlov's theory of classical conditioning showed that behaviour can be learned when a stimulus produces a response, as in his experiment where dogs salivated when they heard a bell associated with food.

Watson applied this idea to humans and argued that Psychology should study observable behaviour, not thoughts.

Skinner later explained operant conditioning, which shows that behaviour can be shaped through rewards and punishments.

Watson VS. Skinner

● Similarities ● Both were behavioral theorists. ● Believed that people were born as blank slates. ● Thought psychology should be studied as though people did not have a mind. ● Wanted scientists to focus on the way people learned connections between stimuli and responses.	● Differences ● Watson was classical conditioning while Skinner was operant conditioning. ● Watson used a human in his experiments while Skinner used animals. ● Watson was the inventor of behaviorism while Skinner just followed in his footsteps. ● Watson had no reinforcements while Skinner did.
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Behaviourism helps to understand how behaviour can be influenced by the environment, habits, and experiences. For example, positive encouragement in families and communities can help people develop good behaviours such as cooperation, discipline, and responsibility. It also helps teachers and parents guide behaviour through reinforcement and good role modeling which supports positive social values in society.

2.2.2. Humanism

Humanistic Psychology developed in the 1950s as a reaction against earlier approaches such as psychoanalysis and behaviourism. Psychologists like Abraham Maslow and Carl Rogers believed these approaches focused too much on negative emotions or learned behaviour and ignored human freedom and personal choice. Humanistic Psychology therefore emphasised that people have free will, personal dignity, and the ability to grow and reach their full potential, a concept known as self-actualisation.

Maslow's Hierarchy of Needs Theory explains that human needs are arranged from basic needs like food and safety to higher needs such as love, respect, and self-actualisation (becoming the best version of oneself). *For example*, a student who is hungry and feels unsafe at home may focus on finding food and security first before worrying about making friends or achieving high grades. Once basic needs are met, the student can seek friendship, recognition from teachers, and eventually work toward personal goals like excelling in sports or music.

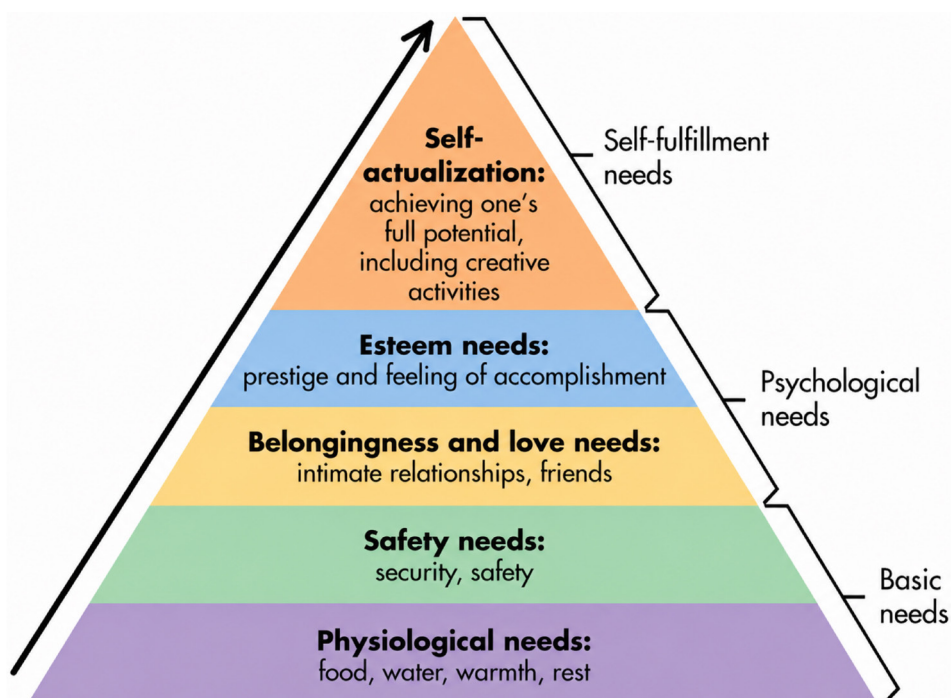
Rogers' theory of client-centred therapy emphasised that people can grow positively when they receive understanding, acceptance, and support from others. *For example*, a teenager struggling with low self-esteem may gain confidence and start expressing themselves creatively when parents, teachers, or counsellors listen without judgment and encourage their strengths.



Carl Rogers (1902–1987)



Abraham Maslow (1908–1970)



This approach helps individuals better understand themselves and develop their abilities. In Psychology, it is applied in counselling and therapy, especially client-centred therapy, where a person is supported in solving personal problems and improving well-being. It also encourages personal development by helping people discover their strengths, follow their goals, and contribute positively to their communities and society.

Humanistic Psychology influenced modern Psychology by promoting a positive view of human nature and focusing on personal growth and happiness. It contributed to areas such as psychotherapy, motivation, and personality, and helped inspire positive Psychology. However, some psychologists criticise it because many of its ideas are difficult to measure scientifically, even though it has helped improve people's quality of life.

Application activity 2.2



1. What is the purpose of the humanistic approach?
2. How do you think humanistic Psychology (self-growth, personal choice, and potential) and behaviourism (rewards, punishments, and reinforcement) can be applied to improve the situation? Consider examples from your school, home, or community.

2.3. Introduction to branches of Psychology: Developmental Psychology, educational Psychology, and cognitive Psychology

Learning activity 2.3



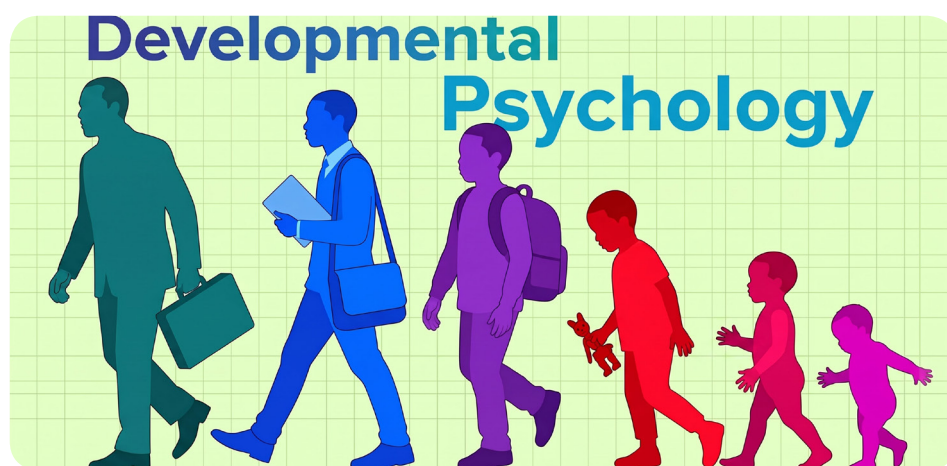
1. Look carefully at the above picture, what is happening in each part of the image? Describe this picture from left to right respecting the sections.
2. Basing on what you see from the picture, what do you know about how we grow and change, think and solve problems, and how we learn?



Image illustrating branches of Psychology

The term “branches of Psychology” means the different areas or parts of Psychology that study different things people do or experience. Each branch looks at a different side of human life, like how people grow and change, how they think and remember, or how they learn at school and in daily life. (Developmental, cognitive, and educational)

2.3.1. Developmental Psychology



Source: [https://www.scientificworldinfo.com/2019/04/developmental-Psychology characteristics-objectives-and-principles.html](https://www.scientificworldinfo.com/2019/04/developmental-Psychology%20characteristics-objectives-and-principles.html)

Developmental Psychology is the study of how people grow, change, and stay the same throughout their lives. It looks at how our thinking, feelings, and behaviour develop from before birth to old age. This includes changes in the body, emotions,

social skills, thinking, and moral understanding. It also considers the influence of biology and culture on how we develop. Jean Piaget, the father of developmental Psychology, began his studies on how children think and grow in the 1920s.

The main goals of developmental Psychology are to describe how people develop, explain why these changes happen, and help people develop better. Psychologists in this field study areas like physical growth, thinking and reasoning, emotions, social skills, and language. They also look at differences between individuals because no two people grow the same way.

When to see a developmental Psychologist?

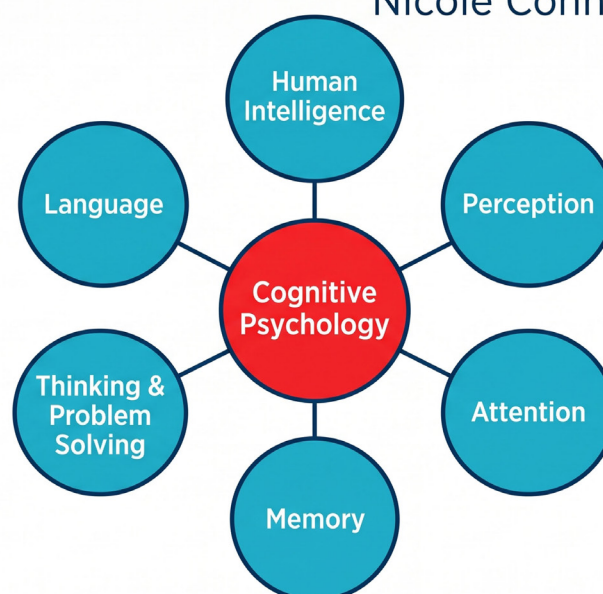
Intervening as soon as possible on behalf of a child demonstrating a delay or challenge in development will significantly improve the issue. At most routine medical checkups, healthcare providers will ask parents about their child's developmental milestones. If the healthcare provider believes milestones are not being reached within a predictable time frame, they may recommend early intervention programs and a more thorough assessment by a developmental psychologist.

The study of developmental Psychology is essential for understanding how humans learn, mature, and adapt throughout their lives as they pass through different stages of development. It helps us understand how people grow physically, emotionally, socially, and cognitively, and how they adjust to changes and challenges at each stage of life. By studying developmental Psychology, we can better support learning, personal growth, and positive adaptation across the lifespan.

2.3.2. Cognitive Psychology

COGNITIVE PSYCHOLOGY

Nicole Conner



Source: <https://www.viquepedia.com/disciplines/cognitive-Psychology>

The father of cognitive Psychology is Ulric Neisser, who is known for starting this field in the 1960s. Cognitive Psychology is the study of how people think and use their minds. It looks at mental processes like attention, memory, perception, problem-

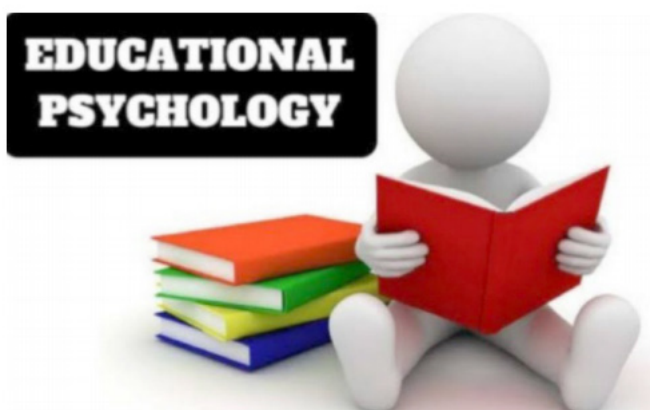
solving, reasoning, language, and creativity. Cognitive psychologists often compare the mind to a computer, where information comes in through the senses, is processed in the brain, and produces responses through decisions, speech, or actions. Some important processes are studied in cognitive Psychology including perception (take information from our senses and make sense of it), memory (store, retain and recall information and experiences), attention (focusing on one thing while ignoring distractions), reasoning (thinking carefully to make decision or resolve a problem), and intelligence (the ability to learn, adapt, and use knowledge effectively).

When to see a cognitive psychologist?

People may see a cognitive psychologist for memory problems, brain injuries, learning difficulties, speech or language issues, or therapy for mental health challenges.

Cognitive Psychology helps us understand how people think, learn, and remember, and it helps develop ways to improve learning and solve cognitive problems.

2.3.3. Educational Psychology



Source: <https://medium.com/@haroon909pk/what-is-educational-Psychology-fb104c976d41>

Educational Psychology is the study of how people learn and grow. It combines Psychology and education to understand how individuals of all abilities learn, what motivates them, and how different methods can improve learning. It looks at both normal learning and difficulties, including challenges like learning disabilities, emotional problems, and special needs in education. This knowledge guides the development of each person's abilities.

Importance of educational Psychology in teaching

A teacher is like a philosopher who guides students on the path of learning and growth. To do this effectively, a teacher must understand the growth and development of each student, which is where educational Psychology becomes essential. It provides teachers with the knowledge and techniques to support learning, motivate students, and channel their abilities in the right direction. By understanding the nature and potentialities of students, teachers can design lessons and activities that encourage students to reach their full potential.

Educational Psychology also helps teachers understand how learning takes place and how the learning process should be initiated. It guides teachers in creating a conducive environment that motivates students, addresses their individual needs, and fosters positive learning experiences. Through this understanding, teachers can adjust their teaching methods according to the students' abilities, interests, and developmental stage, ensuring that students receive instruction that is both effective and engaging. It also equips teachers with the ability to identify and solve learning problems, helping students overcome challenges and progress.

Moreover, educational Psychology is essential for the overall development of a student's personality. It enables teachers to evaluate whether teaching and learning objectives are being met, and to treat students fairly by recognising individual differences. By applying principles of educational Psychology, teachers can guide students in the right direction, support their personal and social growth, and create opportunities for students to develop confidence, skills, and a love for learning.

Importance of educational Psychology in learning

The pioneer of this field, Edward Lee Thorndike, started his work in the 1910s, developing ways to measure intelligence and learning that could improve education and learning in real-life situations.

Educational Psychology helps students in the important task of knowing themselves. By understanding their abilities, capacities, interests, aptitudes, likes and dislikes, attitudes, and dispositions, students can become more aware of their strengths and areas for improvement. This self-awareness allows students to make better choices about how to approach their studies, select suitable learning strategies, and set realistic goals for personal and academic growth.

Educational Psychology provides students with knowledge of learning processes, motivation, memory, and attention. Understanding how to stay focused, avoid distractions, and use effective learning and remembering techniques helps students study more efficiently.

It introduces principles of behaviour modification, which guide students in developing good habits, breaking bad ones, and adopting proper ways of learning. These skills empower students to take responsibility for their own learning and improve their performance over time.

It also helps students understand human development and personality growth. Knowledge of developmental stages and the needs and characteristics of individuals at each stage enables students to recognise their own growth patterns and challenges.

By applying this understanding, students appreciate the importance of all-round development intellectual, emotional, social, and moral which is essential for success in both the processes and outcomes of learning.

Application activity 2.3



1. Describe different changes that a person might be experiencing from 0 to 80 (physical, emotional, social, or intellectual).
2. How can you connect developmental Psychology, cognitive Psychology, and educational Psychology with your life at school, or in your community?
3. As an S4 student who is learning Psychology, how could the knowledge you are learning in Psychology contribute to understanding, guide, or help others learn or do something better in your school, or in your community?

2.4. Branches of Psychology: Social Psychology, Family Psychology

Learning activity 2.4



1. What do you understand by the following terms:
 - a) Family
 - b) Society
2. Based on your experience at home, schools, and community, what do you understand by the words family Psychology, and social Psychology?

Social Psychology and Family Psychology are areas of Psychology that study how people interact with others in their daily lives.

2.4.1. Social Psychology



Source: <https://www.pearson.com/store/p/social-Psychology-seventh-canadian-edition/P100002789163>

Social Psychology is the study of how people's thoughts, feelings, and behaviours are influenced by others and by social situations, focusing on individuals as members of groups or society. It explains how people affect one another through interaction, communication, and relationships, and how culture influences attitudes, obedience, and behaviour toward different groups. Social psychologists study how people learn attitudes, emotions, skills, and ways of living with others, especially through interaction in schools and communities.

The father of social Psychology is Kurt Lewin, who contributed greatly to understanding group behaviour and social influence.

This field is applied in areas such as organising workplaces, reducing conflicts between groups, designing mass communication like advertising, advising on child upbringing, and helping in mental health treatment and rehabilitation, helping people better understand themselves and society.

Social Psychology vs Sociology

Social Psychology and sociology are related fields that study human behaviour in society, but they differ in focus.

Sociology examines the origin, development, structure, and functioning of groups and societies, while social Psychology studies the individual's psychological processes within those groups.

Both fields address issues like violence, prejudice, cultural differences, and marriage, but sociology focuses on group-level patterns, whereas social Psychology focuses on how individuals think, feel, and behave in social contexts.

Importance of social Psychology

Social Psychology is important because it helps us understand how people think, feel, and behave around others. People often act differently in groups than when they are alone, influenced by what others expect or how society behaves. By studying these patterns, social Psychology helps us understand why people make certain choices and how social influence shapes behaviour.

Understanding these social dynamics also improves relationships and reduces problems between people. By learning about empathy, good communication, and the causes of prejudice or stereotypes, individuals can interact more effectively, solve conflicts peacefully, and promote acceptance and fairness in families, schools, workplaces, and communities. This helps people live together more harmoniously.

Moreover, social Psychology encourages positive behaviour and has practical uses in daily life. Knowing how people cooperate, help each other, or respond to social situations can encourage kindness, teamwork, and responsible decision-making. It is also applied in education, marketing, and public policies to create programs that guide people to make better choices and support others. In this way, social Psychology not only explains human behaviour but also helps build stronger and more supportive communities

2.4.2. Family Psychology



Family Psychology studies the emotions, thoughts, and behaviours of individuals within their family relationships, focusing on how family members influence one another. It is based on system theory, which looks at the family as an interconnected unit.

The father of family Psychology is Murray Bowen in the 1950s and 1960s, who developed important ideas about family systems and relationships.

Family psychologists work in hospitals, schools, clinics, businesses, and research centres, helping with family and couple therapy, assessment, consultation, education, and advocacy.

They address issues such as parent-child conflicts, youth behaviour problems, substance abuse, depression, developmental disorders, abuse, intimate partner violence, and general mental health, helping families cope with stress, improve relationships, transform negative behaviour patterns, enhance self-esteem, and stay together for better wellbeing.

Importance of Family Psychology

Improves communication: Family Psychology helps family members understand how to communicate effectively with one another. It teaches them how to express their feelings, listen to others, and share ideas in a respectful way. Good communication reduces misunderstandings and helps family members solve problems together

Strengthens family relationships. People understand the roles, emotions, and expectations within a family. This understanding encourages respect, trust, and cooperation among family members, which leads to stronger and healthier relationships.

Supports child development among families by helping parents understand the emotional, social, and cognitive needs of their children. When parents understand how children grow and learn, they can provide better guidance, support, and care that helps children develop properly.

Helps resolve family conflicts. Conflicts are normal in families, but family Psychology provides ways to manage disagreements peacefully. It helps family members learn problem-solving skills, patience, and understanding so they can resolve conflicts without harming their relationships.

Promotes mental and emotional well-being by creating a supportive and caring family environment. When family members feel loved, respected, and understood, they develop higher self-esteem, emotional stability, and better mental health.

Guides parenting Practices by helping parents to have an understanding of the effective ways of raising their children. It teaches them how their behaviour, discipline methods, and communication style influence their children's behaviour and development. This helps parents raise responsible and well-adjusted children.

Application activity 2.4



1. In your own words, how do you compare and contrast social Psychology, and family Psychology?

2. Through examples, explain how peers and family can influence someone's behaviours.
3. In which ways do psychologists help families in resolving family-based conflicts?

2.5. Introduction to branches of Psychology: clinical Psychology, forensic Psychology

Learning activity 2.5



1. Where do people who have problems with their thoughts, feelings, or behaviour go to get help and treatment?
2. Which organisations help the police and courts investigate crimes and understand people's behaviour to prevent future problems in Rwanda?

2.5.1. Clinical Psychology

Clinical Psychology is the study of understanding and treating people with mental health problems. It was first developed by Lightner Witmer from the USA, who founded the first psychological clinic in 1896. Clinical psychologists assess thoughts, feelings, and behaviours using interviews, observations, and tests to identify causes and plan treatments.



Clinical psychologists also consult with other professionals and organisations and may conduct research to improve mental health care. Their work aims not only to treat disorders but also to prevent problems, support personal development, and help people adapt better to life challenges, promoting overall wellbeing.

When to see a Clinical Psychologist?

Clinical psychologists focus on how behaviour, emotions, and lifestyle affect physical health. They conduct research and provide services such as counselling to help people manage stress, maintain a healthy weight, quit smoking, improve daily nutrition, and reduce risky sexual behaviours.

They also support individuals in hospice care, offer grief counselling, prevent illness, understand the effects of diseases, improve recovery processes, and teach coping skills to handle health challenges more effectively. Their work combines science and practical strategies to promote overall wellbeing.

2.5.2. Forensic Psychology



Forensic Psychology is the study of the connection between human behaviour and the justice system. The field was first developed by Hugo Munsterberg from Germany/ USA in the early 1900s. Forensic psychologists work to improve relationships between individuals and the law by assessing, evaluating, and treating both offenders and victims, and by using Psychology to prevent negative behaviours in the future.

Their main duties include observing and interviewing people in the legal system, writing reports, giving expert testimony in court, providing counseling and therapy, and supervising trainees. Forensic Psychology is important because it helps protect people's rights, prevents repeat offenses, guides investigations, and supports recovery, benefiting both individuals and society.

Importance of forensic Psychology

Forensic Psychology helps us recognize and understand problem behaviors and how they relate to criminal activity, which is an important first step in supporting individuals' recovery. It benefits society by protecting people's rights and preventing repeat offenses, contributing to public safety.

Forensic psychologists assist law enforcement by helping narrow down suspects and focus investigations on those most likely to have committed the crime under investigation.

Application activity 2.5



1. What is the aim of clinical psychologists?
2. What is the main importance of forensic Psychology?
3. When is it advised to see a clinical psychologist?

2.6. Contribution of branches of Psychology to the understanding of human behaviour and language learning

Learning activity 2.6



1. Explain 2 branches of Psychology you know and describe how they contribute to human behaviour.

The branches of Psychology help us to understand how humans think, feel, and behave in everyday life.

They explain why people act in certain ways and how the mind and body work together, which is useful for learning new things, including language.

By studying behaviour, Psychology can help to improve communication, relationships, and social skills, which are important for speaking, listening and understanding others. They show how we learn, remember, solve problems, and use language, which helps in communicating and understanding others.

Having learned about these branches of Psychology also helps us to make good decisions, build confidence, get along with others, and understand ourselves and the world around us.

Overall, Psychology contributes to better decision-making, self-confidence, and understanding of yourself and others, helping people succeed in life and in society.

Application activity 2.6



1. Explain 3 contributions of branches of Psychology to the understanding of human behaviour.
2. How does Psychology help people understand human behaviour?

2.7. End unit assessment



Section A. Knowledge and understanding

1. In your own words, explain the term school of Psychology and branches of Psychology.
2. Identify 2 schools of Psychology and state their main ideas.
3. What is the difference between branches of Psychology and schools of Psychology?
4. Explain the contribution of Sigmund Freud to psychoanalysis.

Section B: Skills

5. Explain how the ideas of behaviourism can help us understand how people learn new behaviour.
6. Describe how developmental Psychology helps explain changes in human behaviour
7. Explain how knowledge from different areas of Psychology can help people understand human behaviour, improve family relationships, and support better learning and communication in everyday life.

Section C: Attitudes and values

8. Explain how knowledge from different areas of Psychology can help people build positive relationships in families and communities
9. Imagine a community is facing problems such as family conflicts, stress, and crime. Propose which branches of Psychology could help solve these problems and explain how

UNIT 3

INTRODUCTION TO HUMAN DEVELOPMENT



Key Unit Competence:

Analyse the physical, cognitive, emotional, and social aspects of human development across the lifespan



Learning objectives

By the end of this unit, you should be able to:

Knowledge and understanding

- Distinguish terms of developmental Psychology
- Outline human developmental domains
- Describe the main domains and periods of human development.
- Explain basic functions of the brain and the role of brain development in influencing learning and behaviour.
- Break down the roles played by the limbic and frontal regions of the brain during emotion processing.

Skills

- Compare and contrast meanings of developmental Psychology terms
- Relate periods of development with those said in Rwandan Culture
- Classify developmental skills according to developmental domains
- Interpret the interrelationships among developmental domains
- Summarise periods of human development based on described behaviours at a given period of time.
- Select some of the social effects resulting from typical emotional processing
- Describe systems that allow the brain to interact with the environment

Attitudes and values

- Appreciate the importance of respecting individual differences in rates and patterns of human development.

- Demonstrate positive attitudes toward supporting others' developmental needs at different life stages
- Value the role of healthy brain development and balanced growth in promoting overall wellbeing and lifelong learning

INTRODUCTION

Have you ever wondered how humans grow and change from infancy to old age? Why do children learn to walk and talk at different ages, or why do people respond differently to similar life experiences? Everyday observations, such as a baby learning to grasp objects, a teenager managing emotions, or an adult adjusting to a new job, show that human development is a lifelong process influenced by physical, cognitive, emotional, and social factors. These experiences remind us that understanding human growth and development requires looking at the whole person across all stages of life.

Human development is the scientific study of how people grow, learn, and change throughout their lifespan. In this unit, you will explore the main domains of human development (physical, cognitive, emotional, and social), and learn how these areas are interconnected. You will also explore developmental principles, milestones, and periods of life, and discover the role of brain development in shaping learning, behaviour, and social interactions.

Through this unit, you will develop the skills to observe, classify, and interpret developmental changes, relate them to your own experiences, and understand the factors that influence development. You will also learn to appreciate individual differences in the rates and patterns of development and recognise the importance of supporting others' development at different life stages.

By the end of the unit, you will be able to analyse human development across the lifespan and explain how brain functioning interacts with environmental and social factors to shape behaviour.

Introductory activity



Arrange the following groups of people according to the Rwandan culture. Do some research and identify the approximate age of each group:

Igikwerere/ijigija, urusoro, igitambambuga, uruhinja, igisekeramwanzi, umwana utashya, inshuke, umusore/inkumi, umwana ukurikira abandi, , ingimbi/umwangavu, igihumuza/gicambyaro, umugabo/umugore, umusaza/umukecuru, ingaragu/umwari.

3.1. Key concepts of human development

Learning activity 3.1



Based on knowledge from the topic of branches of Psychology, select from the following list the terms associated with human development: changes, learning processes, maturation, assessment, growth, development processes, stages of development, child development.

Human development is a branch of Psychology to understand how people grow, develop, and change throughout their lives.

Development refers to the process of change that happens in a person from birth to old age. Development refers to the qualitative changes which include Physical, Cognitive, Social, emotional, and moral changes.

Growth means the quantitative increase in size, length, height and weight. Growth is rapid during the prenatal, neonatal, infancy and adolescent stages and slow during childhood and minimal during adulthood

Maturation refers to the biological growth processes that, relatively uninfluenced by experience, enable orderly changes in behaviour.

Life span refers to the entire duration of a person's life, from conception to death.

Holistic means looking at the whole person, including physical, cognitive, emotional, social, moral and language growth

Reflexes are involuntary, rapid, and automatic movements or actions the body performs in response to a specific stimulus, occurring without conscious thought. Reflexes are primarily protective, designed to keep the body safe from harm (e.g., coughing, blinking, pulling away from a hot stove)

Stimulus is an internal or external actor, event, or situation that elicits or provokes a response from an organism.

Developmental milestones refer to the checklist which represents what an average child can do around a particular age. However, the exact timing of when each child reaches each milestone will vary from one child to another.

Application activity 3.1



1. Using an example, what do you understand by the term "concepts of human development"?
2. Differentiate growth from development

3.2. Principles of human development

Learning activity 3.2



1. Arrange the above pictures from the youngest to oldest child. (Write the number corresponding to the picture).
2. Do you think all children achieve the illustrated skills at the same age? Explain your answer.

The principles of human development are the basic rules that explain how people grow and change throughout life. They show that development is gradual, continuous, and follows a predictable order. These principles also explain that development moves from simple to complex skills, occurs at different rates for individuals, and is influenced by both heredity and the environment. They help us understand the patterns and direction of human growth and development. The main principles of human development include the following:

Development is gradual and continuous: it means that development implies changes that occur in a gradual way; they don't come suddenly.

Development is sequential: it means human development happens in a clear order from one step to another. Example: the child sits down before he crawls; stands before he walks, and babbles before he talks.

Orthogenetic principle: this means that development of more difficult tasks begins with the mastery of simple tasks first. The orthogenetic principle proposed by Heinz Werner states that as a person grows, their thinking becomes more detailed, more organised, and more complex.

Example: A young child may first recognise all animals as simply "dogs" or "big animals" without clear differences. As the child grows, she /he start to distinguish between a dog, a cat, a cow, and a lion based on specific features like size, sound, and behaviour. Later, the child can even classify them into pets, farm animals, and wild animals.

The development occurs at an individual rate (Pace). The rate of development is not uniform, which means that people do not grow and develop at the same speed. Some individuals develop certain skills faster or slower than others.

Development proceeds from general to specific: Development proceeds from general to specific means that a person first shows broad or simple actions, and later develops more detailed and controlled skills

Growth and development are a product of both heredity and environment means that a person develops because of traits inherited from parents (heredity) and experiences from the surroundings such as family, culture, and education (environment).

Cephalocaudal principle (Head-to-Toe) This principle dictates that growth and functional motor control start at the top of the body and move down to the feet

For example: Infants gain control over their neck and head muscles first, enabling them to lift their head, before developing the motor skills needed to sit or walk.

Proximodistal Development (Center-Outward) This principle dictates that growth starts at the centre of the body (trunk) and moves outward toward the extremities (hands and feet)

For example: A baby can move his/her arms (near the trunk) before they can master refined hand movements (far from the trunk).

Application activity 3.2



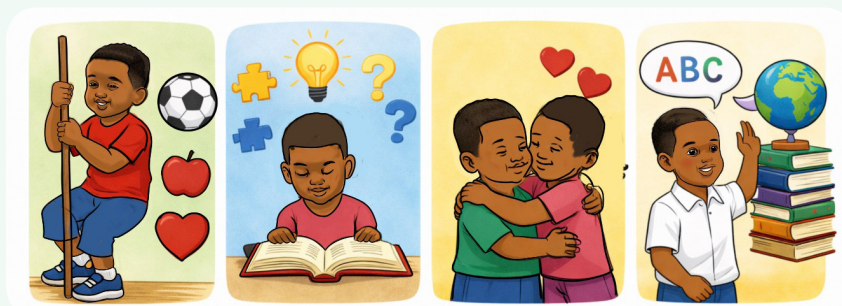
1. Through examples, what does the term “principles of development” mean?
2. Why is it important to know principles of development?
3. Explain these principles using examples:
 - a) Development is gradual and continuous:
 - b) Orthogenetic principle
 - c) The development proceeds from general to specific

3.3. Human developmental domains

Learning activity 3.3



Study the picture below and describe what the children are doing in each section.
How do the actions from the picture connect?



Human development domains refer to the main areas of growth and changes that happen as a person grows from infancy to adulthood. Usually the key domains include:



Image illustrating human development domains

Source: Whole Child Development Research for Inclusive Play (playcore.com)

Physical development and health: involves growing in size and strength, as well as the development of both gross and fine motor skills, brain development, sensory capacities, and health

Cognitive development

Cognitive development refers to the progressive and continuous development of perception, memory, attention, imagination, judgment and reasoning.

Social and emotional development

This domain includes developing attachments to others and learning how to interact and cooperate with them, how to show empathy and use moral reasoning.

Language or communication development

Learning language is the process through which a person learns to understand and use language to express thoughts and feelings. It includes skills such as speaking, listening, reading, and writing.

Developmental areas are connected, meaning growth in one area influences others. For example, physical, cognitive, language, and social-emotional development affect each other throughout a child's growth. Because these areas work together, children need support in all of them to reach their full potential, which is called holistic development. It involves the growth of all interconnected areas of life, promoting balance, resilience, and overall well-being, and helping individuals reach their full potential.

Application activity 3.3



With examples, explain how development in one domain can influence another domain?

3.4. Periods of human development

Learning activity 3.4



1. By looking at the whole collage, can you identify the different stages of life shown from A to E?
2. Arrange these collages from earliest to the latest.

Periods of human development refer to different phases a person goes through from prenatal to old age, each with unique physical, cognitive, social, and emotional changes.

3.4.1. Prenatal period, infancy and toddlerhood, and early and middle childhood

Prenatal period of human development refers to the development in the womb, including growth of organs and body systems. Every person begins life by the union of two sex cells called gametes.

Conception takes place when sperm cell(s) from the male unite with an ovum (egg) in the female's fallopian tube in a process known as fertilisation. By the time the fertilised egg (zygote) ends its three- to four-day journey through the fallopian tube and reaches the uterus, it has divided into about 12-18 cells. A zygote has a full complement of 46 chromosomes. Therefore, the prenatal period is the time from conception to birth.

Prenatal development is divided into 3 main periods: germinal, embryonic and fetal. (More details on prenatal period will be provided in the unit 5).

After conception, the genetic endowment interacts with environmental influences from the start. Basic body structures and organs form; brain growth begins. Physical growth is the most rapid in the life span. Abilities to learn and remember and to respond to sensory stimuli are developing. The fetus responds to mother's voice and develops a preference for it. Vulnerability to environmental influences is great.

Infancy and toddlerhood

- Infancy is a period between 0 and 1 year old. Babies grow rapidly, learn basic motor skills, begin speaking first words, and form strong emotional bonds with caregivers.
- Toddlerhood (1-3years): In this period, children learn to walk and run, expand vocabulary, start basic problem-solving, and explore independence and social interactions.

Early and middle childhood

- Early childhood (3-6 years) is a period in which children develop language, imagination, and social skills. They also improve thinking, problem-solving, and motor abilities through play.
- Middle childhood is a period from 6 to 12 years. In this period, children develop stronger thinking and problem-solving skills, improve reading and writing, and build lasting friendships while refining motor abilities.

3.4.2. Adolescence and early adulthood

Adolescence (12–18years) is the period of rapid physical, emotional, and cognitive changes. During this time, boys may experience a deepening voice, growth of facial hair, get their first nightmares, and increased muscle mass, while girls develop breasts, start menstruation, and gain body fat in typical patterns. Both boys and girls develop abstract thinking, form their identity, and build more complex social relationships.

Early adulthood (19-40 years) is the period when individuals achieve physical maturity, establish careers, form intimate relationships, and may start families. They also develop emotional stability, make long-term life decisions, and refine social and cognitive skills

3.4.3. Middle adulthood and late adulthood

Middle adulthood (41–65 years) is when physical abilities may slowly decline, and women experience menopause. Mental abilities, expertise, and problem-solving peak, while career and personal responsibilities may cause stress.

Late adulthood/Old age (65+ years) is the period when physical strength, stamina, and sensory abilities decline, health risks increase, and individuals may retire. Cognitive abilities may slow, but life experience, wisdom, and emotional maturity continue to grow, while social connections and reflection on life become important. This period consists of the transmission of ideals and family history to the younger generation.

Application activity 3.4



Compare the physical, cognitive, and social changes of males and females across the periods of human development from prenatal to late adulthood. Highlight the differences and similarities at each stage.

3.5. Brain development

Learning activity 3.5



1. What are the main parts or organs found in the human head?
2. We receive information through our senses, but where does the information go for further interpretation?
3. What do you think would happen if the part of the head that controls thinking and body activities was injured?

Every day we think, learn, remember, feel emotions, and make decisions. All these activities are controlled by an important organ in our body called the brain. The brain does not start working fully developed at birth; instead, it grows and changes over time from infancy to adulthood. This process is known as brain development. As the brain develops, it improves our ability to think, learn, control emotions, and interact with others. In this lesson, we will explore the basic structure of the brain and how it develops and functions to control human behaviour and body activities

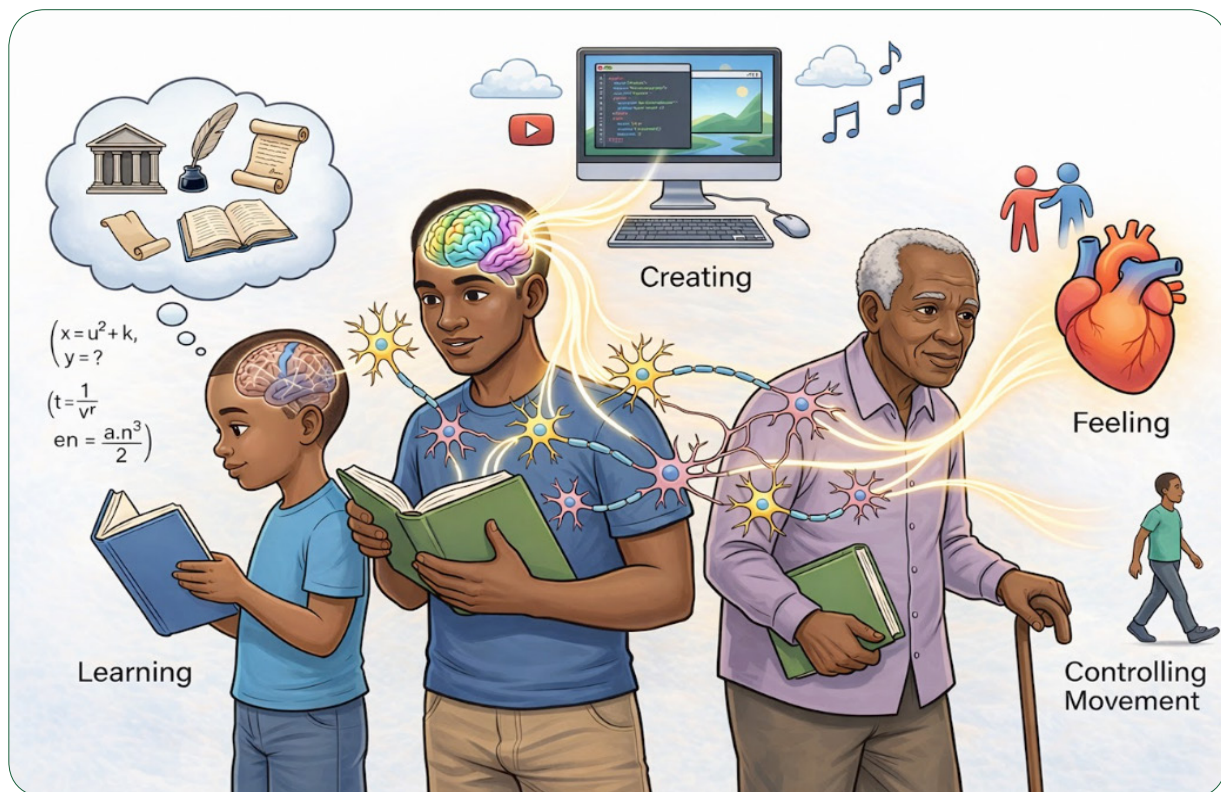


Image illustrating how the brain controls the body's organs and thoughts

The brain receives information through our five senses: sight, smell, touch, taste and hearing, often many at the same time. It arranges the messages in a way that makes sense to us and can store this information in our memory. The brain controls our thoughts, memory and language, the movement of our arms and legs, and the functioning of many of our body's organs.

3.5.1 Parts of the brain and their functions

The brain is composed of the cerebrum, cerebellum and brain stem

- **Cerebrum:** is the largest part of the brain and consists of the right and left hemispheres. It performs higher functions such as interpreting touch, sight and hearing, as well as language, reasoning, emotion, learning and fine control of movement.
- **Cerebellum:** located under the cerebrum. Its function is to coordinate muscle movements, maintain posture and balance.
- **Brain stem:** acts as a relay centre connecting the cerebrum and cerebellum to the spinal cord. It performs many automatic functions, such as breathing, heart rate, body temperature, wake and sleep cycles, digestion, sneezing, coughing, vomiting, and swallowing.

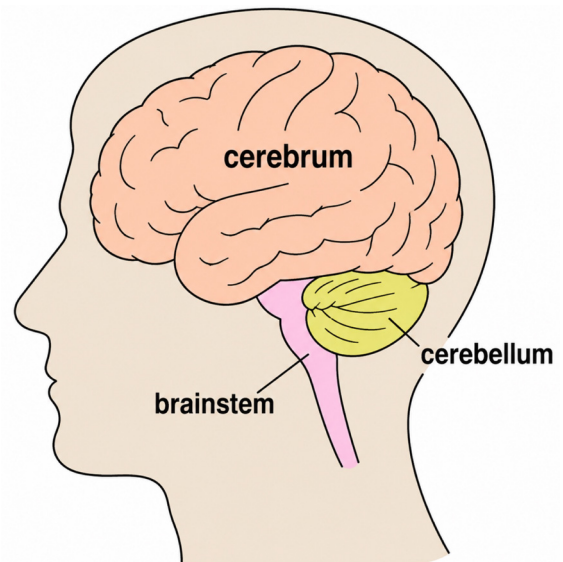


Image illustrating main parts of the brain
Source: <https://mayfieldclinic.com/pe-anatbrain.htm>

Functions of the right and left brain

The left brain and right brain refer to the two halves (hemispheres) of the brain. The human brain is divided into these two sides, which are connected by a bundle of fibres called the corpus callosum, which carries messages from one side to the other.

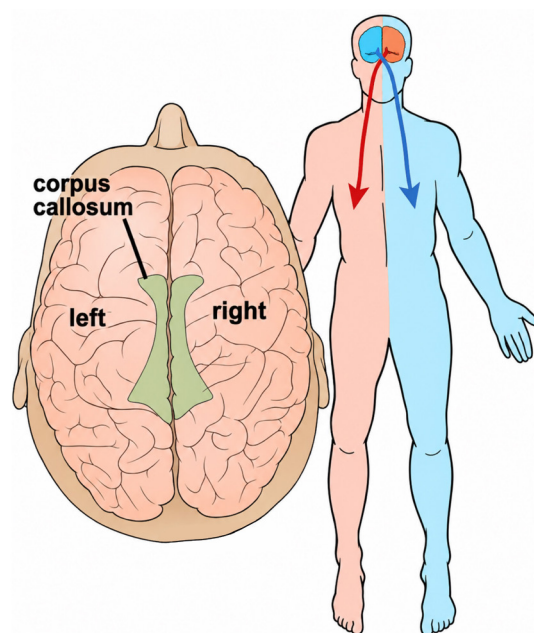


Image illustrating two sides of the brain
Left and right brain Source: <https://mayfieldclinic.com/pe-anatbrain.htm>

In general, the left hemisphere (left brain) is mainly responsible for activities such as language, logical thinking, reasoning, reading, writing, and mathematical skills, while the right hemisphere (right brain) is mainly responsible for creativity, imagination, recognising images, music, emotions, and spatial awareness.

Left-handers face the challenge of being different and a minority (roughly 10% of the population) in a world where almost everything is designed for right-handers. From an early age, they are exposed to everyday situations that require them to make small adjustments, such as opening doors, using tools or scissors. This trains them to use both hands in some activities. However, parents, caregivers, and teachers should not force left-handed children to hold a pencil or pen with their other hand (the right hand). Teachers should consider left-handers so that they have the same opportunities as their peers, other right-handers

Lobes of the brain and their functions

The cerebral hemispheres have distinct fissures that divide the brain into lobes. Each hemisphere has 4 lobes: frontal, temporal, parietal and occipital.

Each lobe, in turn, can be divided into areas that serve specific functions. It is important to understand that not every brain lobe functions on its own. There are very complex relationships between the brain lobes and between the right and left hemispheres.

The frontal lobe, at the front of the brain, is responsible for many important functions. It helps control personality, behaviour, and emotions, allowing us to respond appropriately in different situations. It is also involved in judgment, planning, and problem-solving, helping us make decisions and organise our actions. The frontal lobe controls speech, including both speaking and writing, through a region called Broca's area. It also manages body movements via the motor strips and contributes to intelligence, concentration, and self-awareness, enabling us to think clearly, focus on tasks, and understand ourselves.

The parietal lobe helps the brain interpret language and words. It also processes sensations such as touch, pain, and temperature through sensory strips.

In addition, it interprets signals from vision, hearing, motor activities, sensory inputs, and memory, and plays an important role in spatial and visual perception, helping us understand where things are in space.

The occipital lobe, located at the back of the brain, is mainly responsible for interpreting vision, including colour, light, and movement, allowing us to make sense of what we see.

Temporal lobes, found on the sides of the brain near the ears, help with understanding language, memory, and hearing. They are also important for sequencing and organising information, enabling us to remember events and arrange things in order.

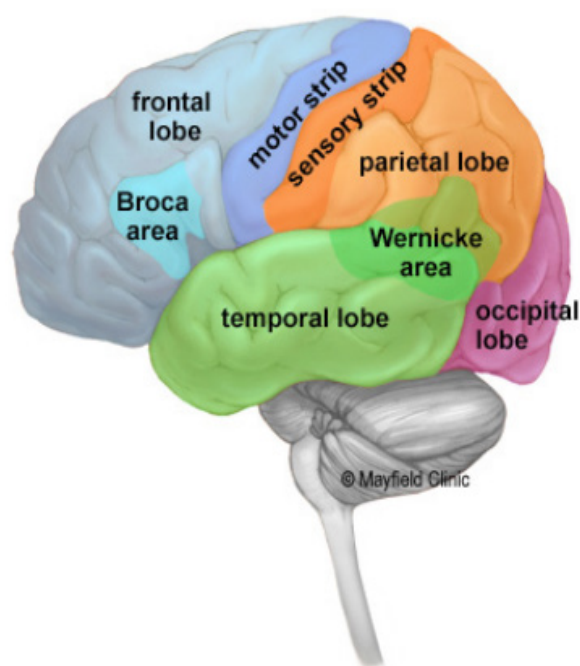


Image illustrating lobes of the brain
<https://mayfieldclinic.com/pe-anatbrain.htm>

3.5.2 Brain development

The brain is the central organ that controls all human development. It influences different domains of development across all stages, which include:

Physical development through controlling body movements, coordination, and motor skills through the motor areas in the frontal and parietal lobes. It helps children learn to crawl, walk, write, and perform other physical activities.

Cognitive development. The brain is responsible for thinking, problem-solving, planning, reasoning, and memory. The frontal lobe, temporal lobes, and parietal lobe all work together to help in thinking clearly, problem-solving, and understanding language.

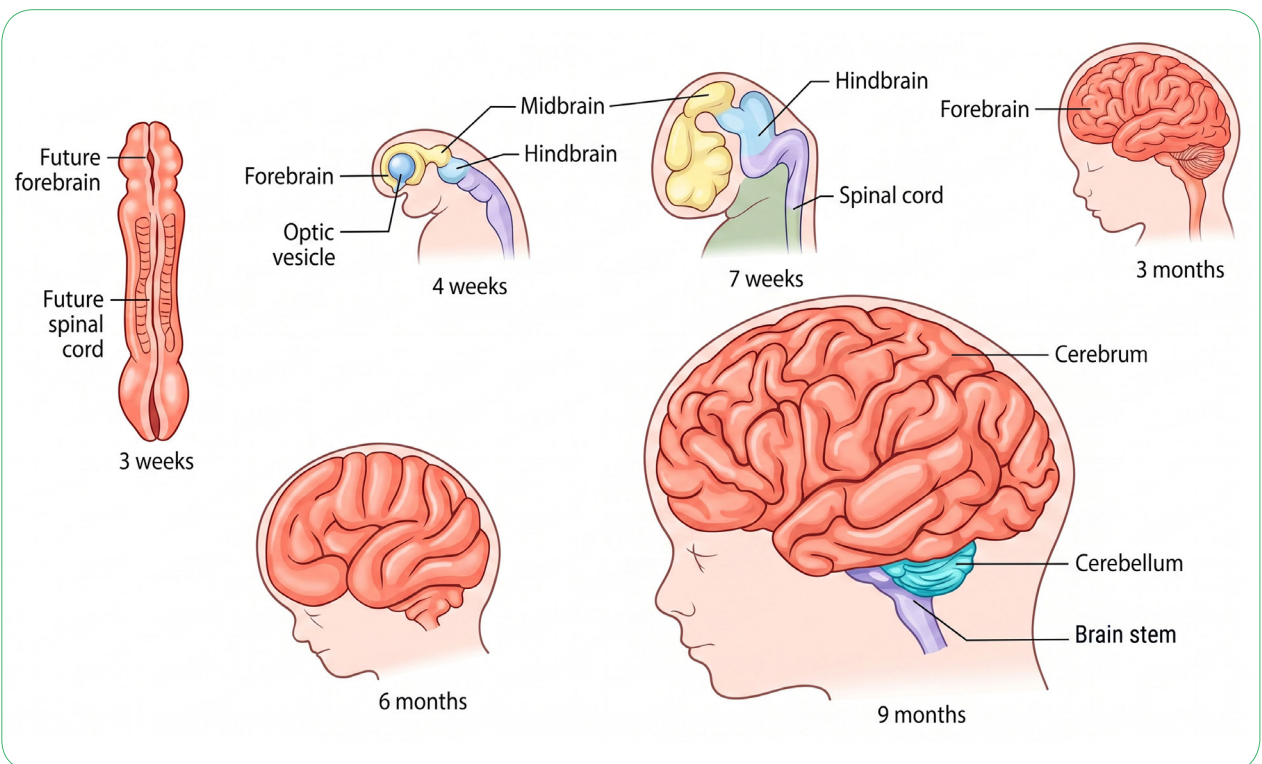
Emotional development by regulating emotions, behaviour, self-awareness, and social interactions. The frontal lobe and limbic system play a key role in helping individuals manage feelings, make decisions, and interact positively with others.

Social development by controlling speech, language, and understanding social cues, the brain helps individuals communicate, cooperate, and build relationships.

Moral development, especially the frontal lobe, helps with judgment, decision-making, and understanding right from wrong, which guides moral behaviour.

The brain begins forming even before birth and continues developing throughout a person's life. During the prenatal stage, the basic structures of the brain form, and neurons start connecting. After birth, the brain grows rapidly, allowing babies to learn, move, and interact with the world. As children grow into infancy, childhood, adolescence, and adulthood, the brain continues to develop, refining skills such as thinking, memory, problem-solving, emotional control, and social interaction. This ongoing development shows that the brain is active and adaptable at every stage of life.

a) Brain development in prenatal period



The development of the foetal brain begins around 3 weeks with the formation and closure of the neural tube, which will become the brain and spinal cord. By 4 weeks, the main regions of the brain appear in a primitive form: the forebrain, midbrain, hindbrain, and optic sacs, which will later form the eyes. As the fetus grows, the forebrain expands, developing into the cerebrum, which controls thinking, memory, and movement. The brainstem, an extension of the spinal cord, is almost fully developed by birth, allowing basic survival functions. The cerebellum, located above the brainstem, grows rapidly during the first year after birth and helps with balance, coordination, and motor skills.

b) Brain development in post-natal period

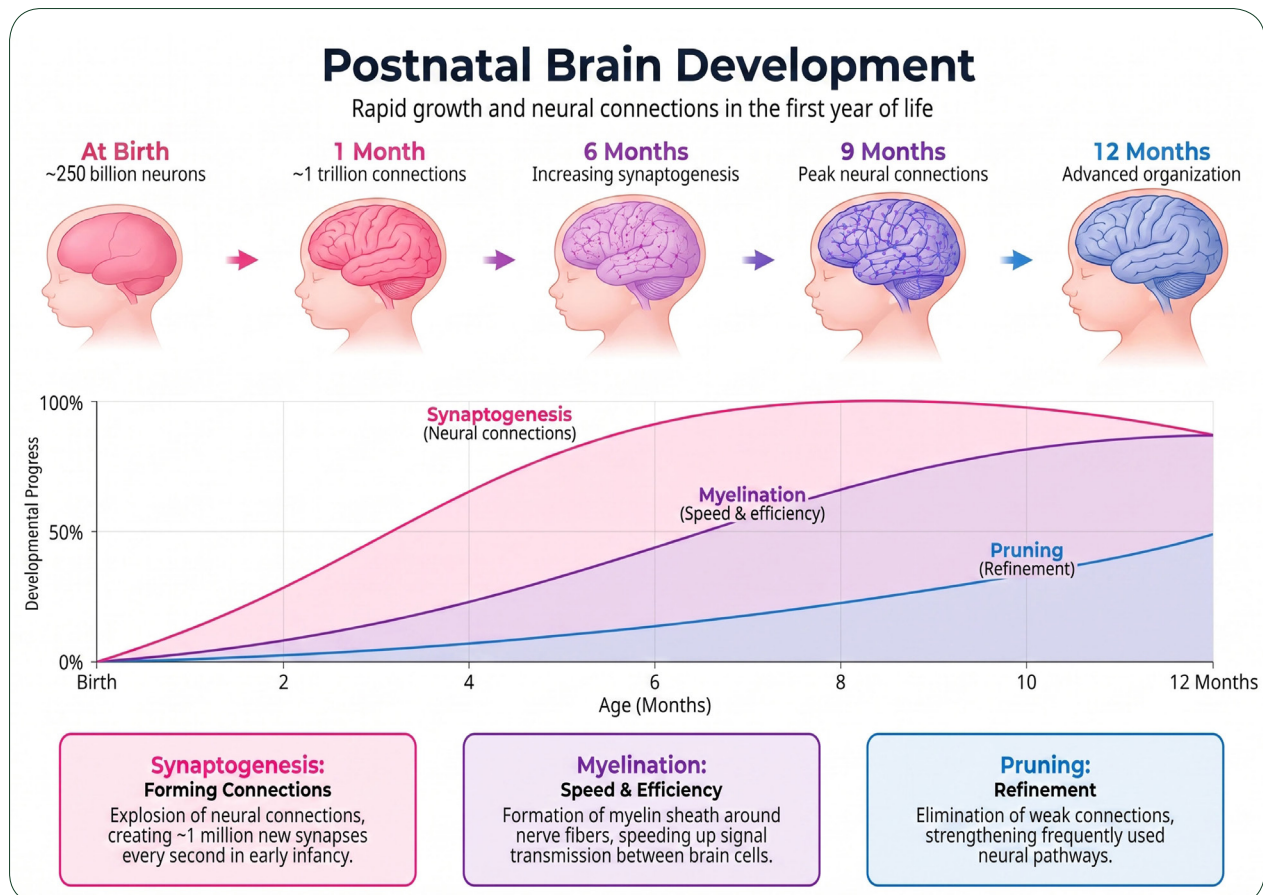


Image illustrating brain development in the post-natal period

At birth, a baby's brain is about a quarter the size of an adult brain, but it grows very fast, doubling in the first year and reaching nearly 90% of adult size by age five. The brain controls everything we do, from moving and talking to feeling emotions, and different areas develop at different rates.

While babies are born with all their brain cells, the real growth happens in the connections between these cells, called synapses.

In early childhood, the brain forms millions of new connections every second, laying the foundation for movement, thinking, communication, and social-emotional skills. These early years are critical because the experiences a child has, positive or negative, strongly shape the brain by helping build important abilities like creativity, problem-solving, self-control, and motivation. Missing these opportunities early makes it much harder to form the same connections later in life.

c) Factors of brain development

From birth to around age five, a child's brain develops more rapidly than at any other time in life. During this period, neurons form billions of connections, creating the foundation for learning, memory, and behaviour. Genetics, nutrition, hormones, and sleep provide the biological framework that supports this growth, while experiences, stimulation, and social interactions shape how these connections form and strengthen. Positive factors like proper nutrition, exercise, and a healthy environment all play a crucial role in supporting optimal brain development.

Brain connections are built through everyday experiences and interactions with trusted caregivers. Loving, responsive care and opportunities to explore the world with the senses help determine which neural pathways are strengthened and maintained throughout life. The quality and consistency of early experiences such as play, communication, and encouragement serve as the building blocks for robust brain architecture, improving learning, behaviour, and long-term physical and mental health.

Application activity 3.5



1. In which period does a child's brain develop more rapidly than at any other time in life?
2. What do you understand by synapses?
3. Identify the 3 main parts of the brain and the main role of each part.

3.6. End unit assessment



Section A: Knowledge and understanding

1. What do you understand by the following terms:
 - a) Human development
 - b) Growth
 - c) Maturation
 - d) Life span
 - e) Holistic development
2. In your own words, explain what human development is and why it matters
3. Name four areas of human development and give one example of a skill attained in each area.
4. Briefly describe three principles of human development.

5. Name three main parts of the brain and describe one function for each.

Section B: Skills

6. Relate the following Rwandan cultural stages to approximate age ranges:

- Igikwerere/ijigija
- Uruhinja
- Umusore/Inkumi
- Umusaza/Umukecuru

7. Explain how physical development in early childhood influences cognitive development.

8. Imagine three children, one learning to draw simple shapes, the second one learning to ride a bicycle, and the third learning to read short sentences. Arrange these from simplest to most complex skill and explain your choice.

9. Give an example of an activity that needs both sides of the brain, and explain what each side does.

Section C: Attitudes and values

10. Why is it important to respect differences in the way children grow and learn

11. Discuss the meaning of the proverb *"Igiti kigororwa kikiri gito"* and describe how it relates to early human brain development and learning.

12. Explain how social media can help spread positive values that support human development in society.

UNIT 4

THEORIES OF HUMAN DEVELOPMENT



Key Unit Competence:

Demonstrate an understanding of theories of human development in real-life

Learning objectives

By the end of this unit, you should be able to:

Knowledge and understanding

- Explain the meaning of concepts necessary to understand the theories of human growth and development.
- Describe theories of human development with their implications for real life
- Describe the main ideas and stages of Freud's, Erikson's, and Piaget's theories of human development.
- Differentiate among psychosexual, psychosocial, and cognitive development theories in terms of their focus and contribution to understanding human growth

Skills

- Compare and contrast theories of human development
- Create scenarios to explain theories of human development
- Sketch stages and main features of the three theories using charts, diagrams, or presentations

Attitudes and values

- Display confidence in commenting about the merits of theories of human development
- Develop positive attitude towards individual's emotions, social, and cognitive differences as explained by the various theories.
- Value the contribution of developmental theorists in improving education, parenting, and personal growth practices.

Introduction

Have you ever wondered why children think differently from adults, why teenagers search for their identity, or how early childhood experiences can influence a person's personality later in life? Psychologists for a long time have tried to answer these questions by studying how people grow and change throughout their lives.

Several psychologists developed theories to explain human development. For example, Sigmund Freud explained development through psychosexual stages, focusing on how early childhood experiences influence personality. Erik Erikson proposed psychosocial stages, emphasising social relationships and identity development. Jean Piaget focused on cognitive development, explaining how children gradually develop thinking and reasoning abilities.

In this unit, you will explore these important theories and learn how they help us understand human growth, behaviour, emotions, and thinking in real-life situations.

This knowledge will also help you better understand your own development and the differences among individuals.

Introductory activity



1. From your previous learning in Introduction to Psychology and Branches of Psychology, what do you remember about Sigmund Freud, Jean Piaget, and Erik Erikson?
2. What do you understand by the term "human development"?
3. A baby cries for milk, a teenager worries about exams, and an elderly person gives advice to younger people.
4. Why do people behave, think, and feel differently at different stages of life? Explain

4.1. Key concepts of human development theories

Learning activity 4.1



1. How do we call the way people change as they grow from infants to adults? Which name can you give to that kind of classification?
2. Recall when you learned something new by connecting it to what you already knew, then write down that situation.
3. What do you understand by the concepts of human development?

Key concepts of human development refer to the most important ideas or terms that help us understand how people grow, change, and learn throughout life.

a) Theory refers to the idea that helps us understand how humans grow, learn, and change,

- b) Stage** refers to a step or period of life where people show certain types of growth,
- c) Erogenous** zone refers to the body areas that play a role in personality development during early childhood (Freud),
- d) Identity formation** refers to the process of learning more about yourself and who you are as you grow,
- e) Assimilation** refers to using what you already know to understand new things,
- f) Cognitive** refers to the process of how people think, learn new things, and solve problems,
- g) Psychosexual** refers to the mental, emotional, and behavioural aspects of sexual development and sexuality, as opposed to purely physical or biological ones
- h) Psychosocial** refers to how our inner mental world and outer social world influence each other and shape well-being, personality, and functioning.

It demonstrates the idea that our relationships with family, friends, and others help shape who we are as we grow.

Application activity 4.1



How do your knowledge and understanding of the key concepts of human development theories contribute to Psychology learning?

4.2. Psychosexual theory (Sigmund Freud)

Learning activity 4.2



1. Reflect on different children from your family or community who are younger than you and identify some common behaviours that change as they grow?

Psychosexual Development Theory is a theory developed by Sigmund Freud and first clearly presented in 1905 in his book *"Three Essays on the Theory of Sexuality"*, which explains how a child's personality develops through 5 stages in early childhood that continue to adolescence.

- 1. The oral stage** is the first year of life; babies get pleasure mainly through the mouth (such as sucking and tasting), and they begin to develop trust in their caregivers who feed and care for them.
- 2. The anal stage** is around 1–3 years old; children learn to control their bathroom habits, which helps them feel independent and responsible depending on how parents/caregivers guide and support them.
- 3. The Phallic stage:** Around 3–6 years old, children become aware of their bodies and the differences between boys and girls. They may feel strong attachment to the parent of the opposite sex and begin to identify with the same-sex parent, which helps them learn about relationships and family roles.

4. **Latency stage.** This stage is from around 6 years to puberty; children focus on learning, friendships, hobbies, and social skills. During this stage, their earlier sexual feelings are less noticeable, and energy is used for school, play, and social activities, which helps them build confidence and relationships.
5. **The genital stage** begins at puberty and continues through adulthood. People develop a strong interest in forming relationships with others and learn to care for others as well as themselves. Completing earlier stages helps them become well-balanced, responsible, and able to manage their feelings and social life.

Application activity 4.2



1. Using an example, explain some 2 stages of psychosexual development theory.
2. Why is Freud's theory called "psychosexual," and what does it explain about personality development?

4.3. Psychosocial development theory (Erick Erikson)

Learning activity 4.3



Reflect on your interactions with schoolmates or community members. How have these experiences influenced your personal growth over a certain period of time?

Psychosocial Development Theory is a theory that describes the process of growing and changing through life by learning from social relationships and experiences with family, friends, and others. It was first published in 1950, when Erik Erikson outlined his ideas about the eight stages of psychosocial development in his influential work "*Childhood and Society*". During each stage, people face a developmental conflict that impacts later functioning and further growth.

1. **Trust vs. Mistrust (0-1)** is the stage where an infant learns to feel safe and confident if caregivers are consistent, or insecure if not, and this later influences old age as either a sense of trust, peace, and security or fear, doubt, and isolation.
2. **Autonomy vs. Shame and doubt (1-2)** At this stage, a child gains independence when supported or feels shame when controlled, leading later to confidence and self-reliance or dependence and insecurity.
3. **Initiative vs. Guilt (3-6)** During this stage, a child develops initiative through encouragement or guilt when discouraged, resulting later in resilience and empathy or lack of confidence.
4. **Industry vs. Inferiority (7-11)** During this stage, a child builds competence through success or feels inferior through failure, leading later to acceptance and humility or feelings of inadequacy.
5. **Identity vs Confusion (12-19)** During this stage, an adolescent forms a clear identity or experiences confusion, resulting later in a strong sense of self or uncertainty about life.

- 6. Intimacy vs. Isolation (20-25):** A young adult forms meaningful relationships or feels isolated, leading later to loving connections or isolation.
- 7. Generativity vs. Stagnation (26-64)** During this stage, an adult contributes to others or feels stagnant, resulting later in care and concern for others or lack of purpose.
- 8. Integrity vs. Despair (65 +)** At this stage, an older adult reflects with satisfaction or regret, leading to wisdom and peace or despair and dissatisfaction.

Table summary of stages of Psychosocial development

#	Age range	Conflict	Resolution/ Virtue	Outcome or Final result in old age
1	0 – 1	Trust vs mistrust	Hope	Appreciation of interdependence and relatedness
2	1 – 2	Autonomy vs shame and doubt	Will	Appreciation of interdependence from integration to disintegration
3	3 – 6	Initiative vs guilt	Purpose	Humour, empathy, resilience
4	7- 11	Industry vs inferiority	Competence	Humility, acceptance of the course of one's life and unfulfilled hopes
5	12 - 19	Identity vs confusion	Fidelity	Sense of complexity of life, merging of sensory, logical and aesthetic perception
6	20 - 25	Intimacy vs isolation	Love	Sense of the complexity of relationships, value of tenderness and loving freely
7	26 - 64	Generativity vs stagnation	Care	Caritas - caring for others, and agape - empathy, and concern
8	65 -	Integrity vs Despair	Wisdom	Existential identity; a sense of integrity strong enough to withstand physical disintegration

Application activity 4.3



- Using examples, explain some 2 Erickson's stages of human development and describe how experiences in each stage can influence behaviour later in life.
- What factors do you think cause some people to marry in their 20s while others may choose not to marry at all?

4.4. Cognitive development theory (Jean Piaget)

Learning activity 4.4



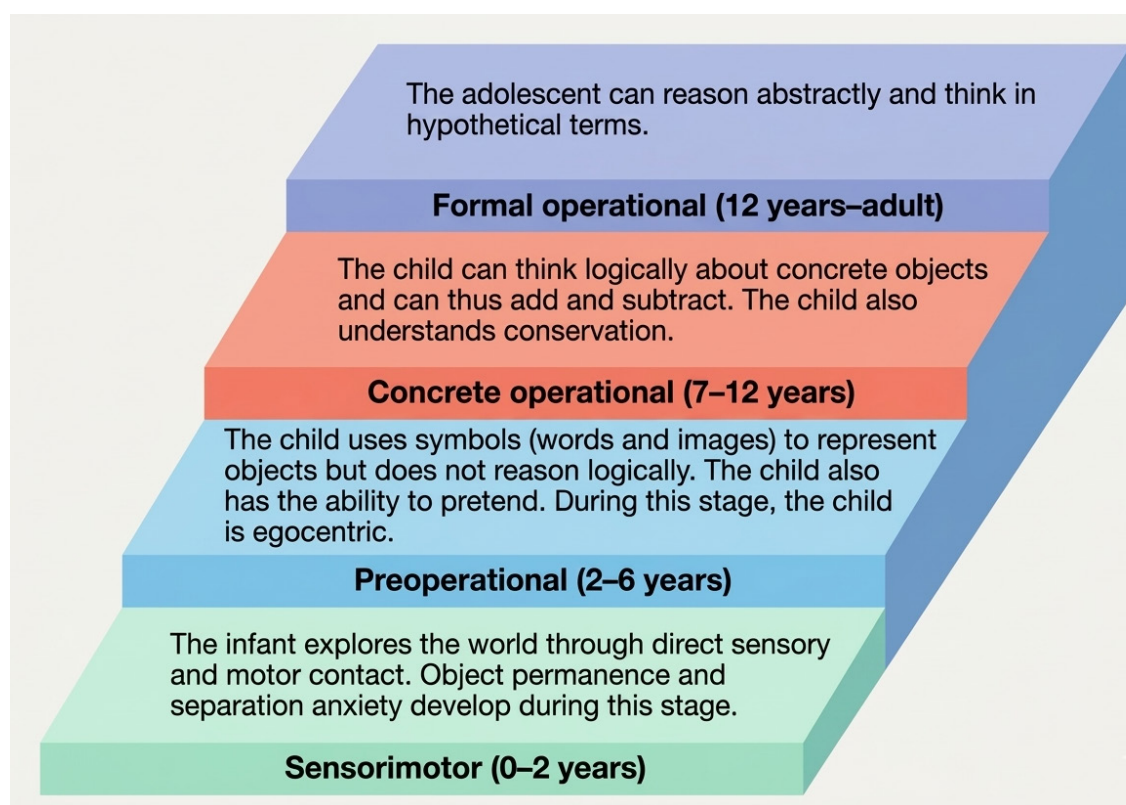
1. How do you understand the concept "cognitive Psychology"?
2. What do you know about Jean Piaget?

Cognitive Theory of Human Development studies how a person's thinking develops and affects how we understand and interact with the world. Jean Piaget, a key theorist, showed that *children think differently from adults*. Through observing his own children, Piaget proposed a stage theory of intellectual development with four distinct stages, describing how children's thought processes grow and change over time.

Cognitive theory includes: mental processes, knowing and understanding, memory, thinking, attention, problem solving, and language.

For example, when solving an equation in mathematics, we use memory, reasoning, and attention.

The stages of cognitive development



Source: https://www.researchgate.net/figure/Piagets-Stages-of-Cognitive-Development_fig12_328788893

1. **Sensorimotor (Birth–2 years)**: This stage was explained by Jean Piaget. During this stage, babies learn about the world through their **senses and physical actions**. Babies learn through senses and actions, like shaking a rattle to hear the sound, and know things exist even when out of sight.

Characteristics of the Sensorimotor Stage of cognitive development

- ❖ **Learning through senses and movement:** Babies explore the world by touching, seeing, hearing, and moving. Example: A baby touches a blanket to feel its softness.
- ❖ **Using basic actions like sucking, grasping, looking, and listening:** Example: A baby puts a toy in the mouth or holds a parent's finger.
- ❖ **Understanding object permanence (things exist even when not seen):** Example: A baby looks for a toy hidden under a cloth.
- ❖ **Realising they are separate from other people and objects:** Example: A baby moves their hand to grab a toy.
- ❖ **Learning that their actions cause things to happen:** Example: A baby shakes a rattle and hears a sound.

2. Preoperational (2–6 years): Children use words and pictures but see the world mostly from their own view, like thinking everyone knows the game they are playing.

Major Characteristics and Developmental Changes

- ❖ Children begin to think symbolically and learn to use words and pictures to represent objects. A child draws a cow and says it is "Inka yanjye" (my cow), or uses a stick as a pretend hoe while playing farmer.
- ❖ Children at this stage tend to be egocentric and struggle to see things from the perspective of others. A child might cover their eyes and think that if they cannot see someone, that person also cannot see them.
- ❖ While they are getting better with language and thinking, they still tend to think about things in very concrete terms. A child may believe that if water is poured from a short, wide cup into a tall thin cup, the tall cup has more water because it looks higher, even though the amount is the same.

3. Concrete Operational (6–11 years): Children think logically about real things, like knowing that water in a tall glass is the same amount as in a short glass.

Major characteristics and developmental changes

- ❖ During this stage, children begin to think logically about concrete events. A child understands that if water is boiled on the stove, it becomes hot and can burn.
- ❖ They begin to understand the concept of conservation; for example, that the amount of liquid in a short, wide cup is equal to that in a tall, skinny glass,
- ❖ Their thinking becomes more logical and organised, but still very concrete. A child can sort objects by size or colour, such as arranging mangoes from smallest to largest.
- ❖ Children begin using inductive logic, or reasoning from specific information to a general principle. After seeing that one banana and one orange are sweet, a child might conclude that all fruits are sweet

4. Formal Operational (12 years and up): Teens and adults can think about ideas that are not real or visible, imagine different possibilities, and solve problems step by step.

Major Characteristics and Developmental Changes

- ❖ At this stage, the adolescent or young adult begins to think abstractly and reason about hypothetical problems. Abstract thought emerges. A teenager can imagine what might happen if they or their friends engaged in sexual activity early, such as the possibility of getting pregnant, contracting a disease, or facing social and emotional problems, even though it hasn't happened yet.
- ❖ Teens begin to think more about moral, philosophical, ethical, social, and political issues that require theoretical and abstract reasoning. A young adult debates whether it's fair to enforce strict laws on environmental protection, considering both economic and ethical perspectives
- ❖ Begin to use deductive logic, or reasoning from a general principle to specific information. For example, if a teen knows that "all humans need oxygen to survive" (general principle), they can deduce that "if someone is underwater without air, they will not survive" (specific case).

Application activity 4.4



Read the story "*Play time*" and do the task.

It is play time to Mahoro' house, her children are playing in different corners, 1-year old boy is playing with toys by shaking, touching, and putting them in his mouth, her 4 years old daughter is using blocks to build a "castle" and explaining the story behind it, her 7 years old son is measuring water from one cup to another to see which has more, and the 13 years old daughter is doing the homework imagining different ways to solve an equation.

Task:

- a) Identify and describe the stage of cognitive development of each child is in, and
- b) Explain how this knowledge can help in improving teaching and parenting.

4.5. End unit assessment



Section A: Knowledge and understanding

1. What do you understand by the following concepts:
 - a) Cognitive
 - b) Psychosocial
 - c) Psychosexual
2. State the stages of the theory of human development (choose Freud, Erikson, or Piaget) and explain how they influence human behaviour.

Section B: Skills

3. Read the situation below and answer the questions:

A young child is playing with toys by putting them in their mouth, but a sibling takes one away because it is sharp. Another child brings colourful toys that are safe to play with, while their older sister stands nearby, watching and waiting for them to finish and tidy up the play area.

Question: Identify the stage of development for each child and explain your answer.

4. Create a short scenario or story that shows one stage from any theory of human development.

Section C: Attitudes and values

5. Can knowledge of human development theories help you to develop a positive attitude towards others' emotions, social, and cognitive differences
6. Among the three theories of human development, which one do you find most useful and why?

UNIT 5

INTRODUCTION TO DEVELOPMENTAL MILESTONES



Key Unit Competence:

Analyse human developmental milestones across the lifespan based on physical, cognitive, language and socio-emotional skills.

Learning objectives

By the end of this unit, you should be able to:

Knowledge and understanding

- Explain the key concepts related to developmental milestones
- Describe what happens during prenatal.
- Identify the risk factors for women and infants during pregnancy and delivery
- Explain the developmental milestones for each stage of development

Skills

- Classify milestones for each period of development
- Arrange prenatal developmental milestones
- Create scenarios sequencing human developmental milestones from birth to adulthood.
- Make a linkage supporting activities to the developmental milestones for each domain of development.
- Develop a key message to pregnant ladies

Attitudes and values

- Appreciate the role of parents in supporting their child's development
- Value the importance of prenatal care to support development
- Appreciate that every stage of human development has positive and critical aspects.

Introduction

Have you ever thought about how people grow and change from the moment they are conceived to old age? Every stage of life prenatal, childhood, adolescence, adulthood, and old age comes with unique physical, cognitive, language, and socio-emotional milestones. These milestones are important because they show how humans develop skills, adapt to their environment, and interact with others.

In this unit, you will explore human developmental milestones across the lifespan. You will learn what happens during prenatal development, the key skills children acquire as they grow, and how adolescents and adults continue to develop in different areas. You will also examine the risk factors that can affect development, especially for women and infants during pregnancy and delivery.

Through different engaging activities, you will have practices of classifying and linking developmental milestones to real-life situations. You will develop messages to educate pregnant women and reflect on the important role parents and caregivers play in supporting individual development. You will also learn about the effects of the digital/technology world on human development, especially among youth. You will explore how using phones, the internet, social media, and other digital tools can positively influence learning, thinking, social skills, and behaviour, as well as potential negative impacts on development.

By the end of this unit, you will be able to explain human developmental milestones, identify ways to promote growth in each domain, and appreciate the positive and critical aspects of every stage of life. This knowledge will help you understand yourself, others, and the importance of nurturing development at every age

Introductory activity



Think about a baby, a child, a teenager, and an adult in your community,

1. What can toddlers do that they could not do at birth?
2. What new abilities do teenagers develop compared to younger children?
3. Which difference do you notice in the way adults behave compared to children and people of young ages?
4. What behavioural changes do you observe among people at different ages in your community, and what challenges do they face at each stage of life?

5.1. Key concepts related to developmental milestones

Learning activity 5.1



Based on what you learned from the previous unit,

1. What do you understand from the following concepts:

Conception

Childhood

Prenatal

Adolescence

2. How can you differentiate an adult from old age?

Key concepts of developmental milestones are the main ideas and terms that help you understand the important skills children achieve as they grow and the stages at which these skills develop.

- **Milestone:** a significant stage in someone's development
- **Conception** means the beginning of life, when a sperm fertilises an egg.
- **Prenatal** refers to the stage before birth, when the baby grows and develops inside the mother's womb.
- **Childhood** refers to the stage that involves rapid growth where children learn to speak, play, think, and interact with others.
- **Adolescence** is the teenage stage, marked by physical changes, emotional development, and more independent thinking.
- **Adult** refers to a fully grown person who has reached maturity and is responsible for their own decisions and actions
- **Old age** is the later stage of life, where people may slow down physically but have more experience and wisdom.

Application activity 5.1



1. Through a scenario, explain the key concepts of developmental milestones.
2. How can you differentiate a mature person from an old-age person?

5.2. Prenatal developmental milestones

Learning activity 5.2



1. Look carefully at the image above and connect it to when you saw a pregnant woman in your village. What changes do you notice in that woman?

2. Why do you think the child on 9 months rotate downward inside the mother's womb?

Prenatal development is the period from conception to birth, lasting about 9 months. During this time, a single-celled zygote develops in a series of stages into a full-term baby. The three primary stages of prenatal development are the germinal stage, the embryonic stage, and the fetal stage.

Timeline of Life Before Birth

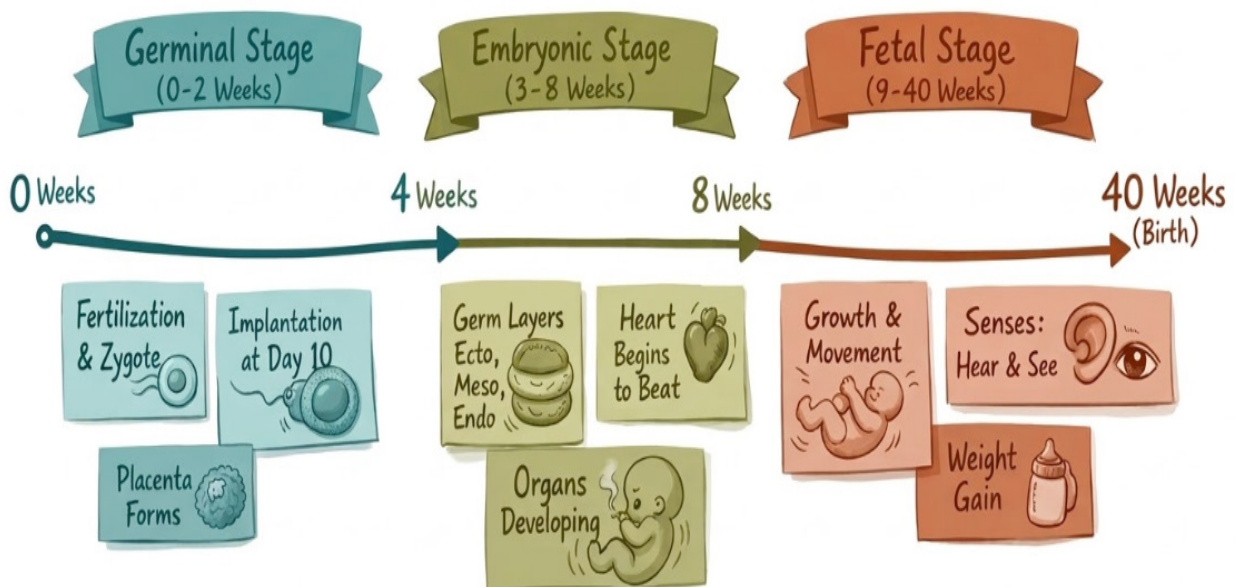
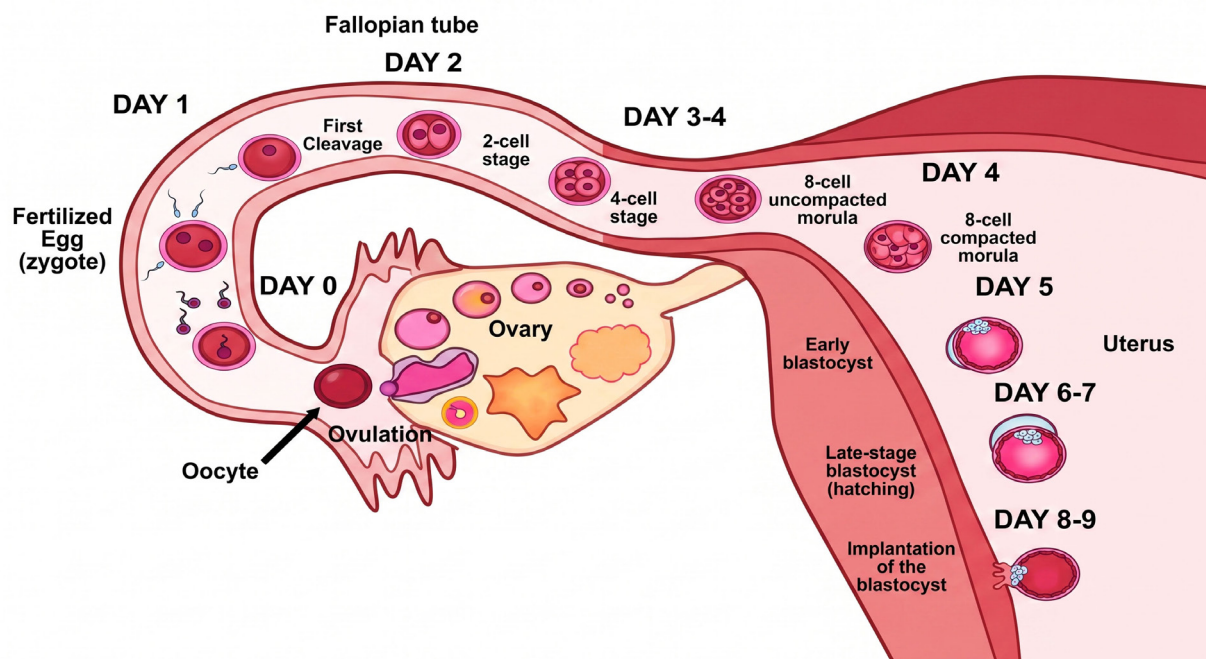


Image illustrating timeline of life before birth

Germinal stage

The first and shortest stage of human prenatal development, lasting approximately 10 to 14 days from conception to implantation. It begins when a sperm fertilises an egg to form a zygote in the Fallopian tube, which then undergoes rapid mitosis (cleavage) while travelling to the uterus, transforming into a morula and then a blastocyst. The stage concludes with the blastocyst implanting into the uterine wall.



Source: Human_Growth_and_Development%2F23.2%253A_Germinal_Stage=AOvVaw3xL_2qP9bMWJq6OP-27bxvY=

During the germinal stage, the zygotic period occurs in the first 2 weeks 1–2: The zygote forms when sperm and egg unite, divides rapidly, and develops inner and outer layers. By day 10, implantation occurs in the uterus. The placenta, umbilical cord, and amniotic sac begin forming to support the fetus.

Human Embryonic stage

The **embryonic stage (weeks 3–8)** is a critical period of prenatal development that begins after implantation. **This period is characterized by** rapid differentiation of the embryo and the formation of three germ layers in week 3: the **ectoderm**, which develops into the nervous system, brain, spinal cord, and skin; the **mesoderm**, which forms muscles, bones, the heart, and other internal organs; and the **endoderm**, which gives rise to the digestive system, lungs, and other internal linings.



From weeks 4 to 8, organogenesis occurs, during which major organs and structures such as the brain, spinal cord, heart, eyes, ears, and limb buds begin to develop. The heart starts beating around weeks 5–6. This stage is highly sensitive to environmental factors and is foundational for structural development.

Foetal period

The **foetal stage (weeks 9–40)** is the longest period of prenatal development, focused on rapid growth, organ maturation, and functional refinement of systems formed during the embryonic stage.

This period is characterised by significant increases in size and weight, development of muscles and coordinated movements, maturation of organs like the lungs, brain, and liver, and functional senses such as hearing, sight, and taste. Facial features become more defined, and hair (lanugo) and skin develop as the fetus prepares for birth.

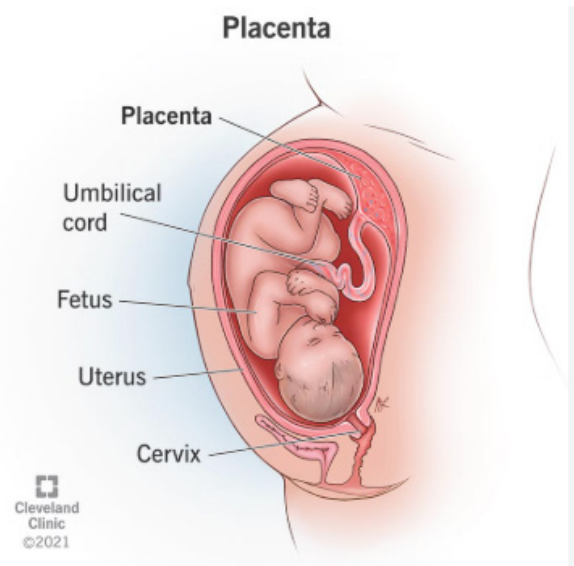


Image of the whole prenatal period



Application activity 5.2



Fill in the table below with the key milestones for each prenatal development stage (Germinal, Embryonic, Fetal) and write a short explanation of why each milestone is important.

Stage	Weeks	Key event/Milestone	Notes/explanation

5.3. Risk factors for women and infants during pregnancy and delivery

Learning activity 5.3



Kanyana is an expectant mother. She is not good at home since she is living in extreme poverty. Her husband always comes home drunk and beats her a lot. Kanyana is suffering from malnutrition and an intense fatigue.

1. Do you think Kanyana's life experience will impact her baby?
2. Explain how and why?

Risk factors for women and infants during pregnancy and delivery refer to conditions or behaviours that increase the risk of complications during pregnancy. There are several factors, influences or hazards that can harm an unborn child during the prenatal period or during birth.

5.3.1. Prenatal risk factors (during pregnancy)

Maternal illnesses such as diabetes and infections like rubella, syphilis, or HIV can harm the baby and may cause physical or mental problems. Some infections can pass from the mother to the baby during pregnancy or at birth. Poor maternal health, including stress and lack of proper nutrition, can also affect the baby's brain and body development.

The age of the mother is important, as very young or older mothers (above 40 years) have a higher chance of complications. Blood incompatibility (Rhesus factor) between mother and baby can also create problems, especially in later pregnancies.

Environmental factors like radiation, pollution, and toxic chemicals can damage the fetus. Similarly, psychological stress can affect the baby by reducing oxygen supply. The use of harmful substances such as alcohol, cigarettes, and drugs can lead to birth defects, low birth weight, or even death.

Miscarriage may occur due to infections, physical problems, or trauma.

5.3.2. Risk factors during delivery

For most women, the birth process, as painful as it may be, proceeds normally. Occasionally, however, problems arise. The following are a few of the more common complications:

Forceps delivery is used to help pull the baby out when the mother is too tired or there are problems during labour, though it has small risks.

Breech birth occurs when the baby is not in a head-down position, making delivery more difficult.

A cesarean section is a surgical method used when normal delivery is not possible to ensure the safety of the mother and baby.

Application activity 5.3



1. Describe risks or influences that are most likely to affect women and infants during pregnancy and delivery in your community and suggest how they can be managed/ prevented.
2. Explain how age can affect both the mother and the infant during prenatal.

5.4. Key message and recommended practices during 1000 Days

Learning activity 5.4



Reflecting on risks affecting women and children before and after birth in your community, suggest 3 Pieces of advice that can help to handle these challenges.

From pregnancy until a child's second birthday, the first 1,000 days offer a unique window of opportunity to build healthier and more prosperous futures.



a) During pregnancy, birth and thereafter confirming pregnancy,

The mother should attend regular antenatal care (ANC) visits to monitor her health and the baby's development each trimester (*1st trimester: 0- 3 months, second: 4-6, third: 7-9*). She should receive proper information about pregnancy, vaccinations, and safe delivery, and be encouraged to test for HIV to protect both herself and the baby. Maintaining a balanced diet is essential for good health, while avoiding alcohol, drugs, and smoking is important to prevent harm to the unborn child. Additionally, any medication should only be taken after consulting a health worker.

b) During birth and after birth (0-2years)

During birth, the mother should deliver at an authorised health care facility. Immediately after birth, the baby should be kept warm, clean, and safe, and breastfeeding should begin within the first hour. Both the mother and baby should be closely monitored to quickly identify and manage any complications.

c) From birth to 6 months,

Parents should practice exclusive breastfeeding, follow the baby's immunisation schedule, maintain good hygiene, and offer love, warmth, and responsive care while monitoring growth and attending health check-ups. They should also support the mother's recovery during this period.

d) From 6 months to 2 years,

The introduction of complementary foods should be at 6 months. Parents should ensure hygiene, complete all immunisations, encourage play, communication, and learning, and provide a safe and supportive environment for the child's overall development.

Effect of Health, hygiene, balanced diet and rest on the pregnant mother

If a mother does not follow these healthy practices during pregnancy, it can lead to serious effects for both her and the baby, such as:

For the baby: miscarriage, premature birth, low birth weight, birth defects, infections, or being born addicted to harmful substances.

For the mother: complications during pregnancy or delivery, poor health, increased stress, and difficulty recovering after birth.

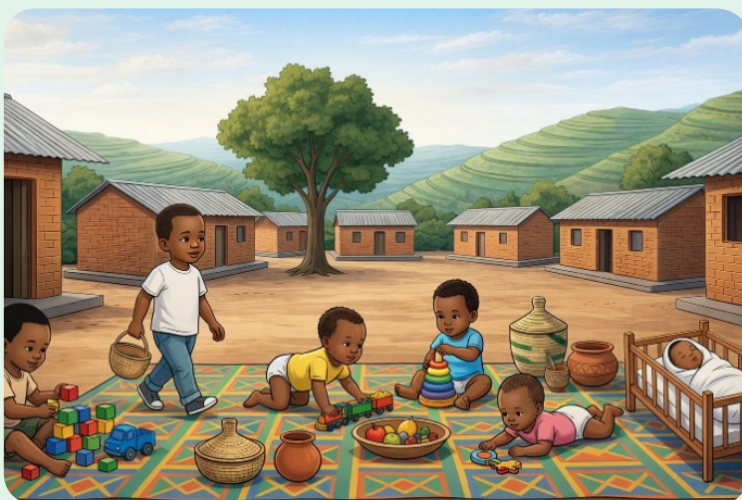
Application activity 5.4



1. How can you advise a woman who is in the 2nd trimester of pregnancy?
2. Why do both mother and child have to be vaccinated?

5.5. Developmental milestones during infancy (0-2years)

Learning activity 5.5



1. Which age range are the children in the image?
2. What are the physical characteristics of children in the age range of 0-6 months, 6-1 year, and 1-2 years?

Infancy is the period of a child's life from birth up to 2 years old. During this period, children show rapid growth at different stages.

5.5.1. Developmental milestones during (0-6)

During the first six months, infants grow quickly in all areas.

Physically, they lift their head, roll over, grab toys, and bring their hands to their mouth.

Cognitively, they explore objects and recognise familiar people. **Communication and language** develop as they make sounds, smile, and respond when someone talks to them.

Social and emotional skills grow as they enjoy playing with people and respond to feelings. Parents can help by playing, singing, talking, reading, giving safe toys, and holding the baby close.

5.5.2. Developmental milestones that babies reach during 7-12 months

During this period, infants grow in all areas:

Physically, they sit without support, crawl, pull themselves to stand, and pick up small objects.

Cognitively, they explore toys, find hidden objects, and copy gestures. **Communication and language** develop as they say simple words, use sounds with gestures, and follow easy instructions.

Social and emotional skills grow as they enjoy playing with people, show preferences for certain toys or people, and test reactions. Parents can help by playing with them, reading, talking, and giving safe toys to explore.

5.5.3. Developmental milestones during 13-24 months

During this period, toddlers grow quickly in all areas.

Physically, they walk, run, climb, kick balls, and use their hands for feeding, drawing, and stacking blocks.

Cognitively, they explore toys in different ways, solve simple problems, and start pretend play.

Communication and language develop as they use more words, make short sentences, and understand simple instructions.

Social and emotional skills grow as they play with other children, show independence, follow simple rules, and begin to understand feelings. Parents can support them by encouraging active play, reading, talking, and helping them join daily activities safely.

Application activity 5.5



1. Which social skill does 0-6month infant develop?
2. Explain one way to help a 13-24-month-old develop communication and language skills?
3. Give two ways parents or caregivers can support an infant's development in reaching their milestones.

5.6. Developmental milestones during childhood

Learning activity 5.6



1. Look at the children in the picture; which age range do you think they are?
2. Compare them with the ones from your village. What behaviours do they display by age?

Developmental milestones during childhood are key skills that most children reach by specific ages. These milestones are essential for monitoring healthy growth of a child during early childhood and middle childhood, and the following are the key areas of development:

Motor (Physical): Gross motor skills (walking, jumping) and **Fine motor** skills (picking up food, drawing).

Language/Communication: Understanding language and expressing needs through words or gestures.

Social and Emotional: Understanding emotions, playing with others, and showing affection.

Cognitive: Learning, thinking, and problem-solving

5.6.1. Developmental milestones during early childhood (3-6years)

During this stage, children stop being toddlers and start becoming young students (umwana ukurikira abandi). They can do many things on their own and become much more interested in the world around them. They become stronger, able to run and climb easily, and their hands become skillful enough to draw shapes and use scissors. This is a time of big imagination and many "Why?" questions as they learn to recognise colours, numbers, and basic ideas about time. Socially, they move from playing alone to cooperating with friends, learning to share, and understanding their own emotions as they take pride in doing things independently.

5.6.2. Developmental milestones during middle and late childhood (7-12years)

During this stage, children grow significantly in their thinking, physical abilities, and relationships, becoming more independent and focused on school and friends (umwana utashya/utumwa). They become more coordinated and skillful, allowing them to participate in activities like sports, music, and typing. They learn to think logically, understand how things work, and understand concepts like time and numbers more clearly. Friendships become very important, and children learn to work together,

solve problems with others, and understand how people feel. This is also a time when children begin to understand who they are and their place in the world, while facing new social challenges.

Application activity 5.6



1. What is childhood?
2. How do key development domains affect each other during childhood development milestones?
3. What are the similarities and differences between early childhood and middle childhood in childhood developmental milestones?

5.7. Developmental milestones during adolescence (11-22)

Learning activity 5.7



1. What do you know about adolescence?
2. Which physical changes do you notice in a 14-year-old boy and girl?
3. What are behaviours that most of people display in their 16s?

The term adolescence is derived from a Latin word 'adolescere' that means to grow to adulthood. Adolescence begins as a separate developmental stage at about the age of 11 and ends at about the age of 21. Adolescence is divided into three stages, which are early, middle, and late adolescence, each with physical, cognitive, emotional, and social changes.

5.7.1. Developmental milestones during early adolescence (11-13years)

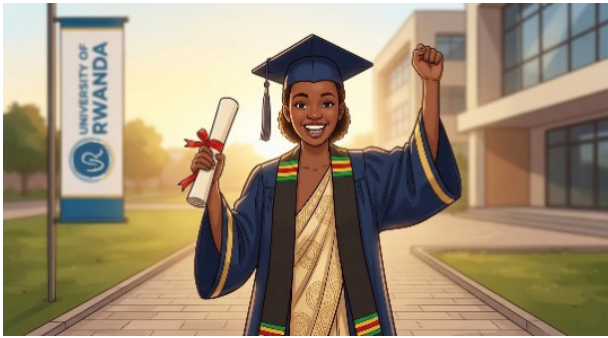
This is the stage where puberty begins, causing rapid physical growth and body changes. Thinking is simple and self-centred, while emotions include mood changes and sensitivity about appearance. Peer approval becomes important, with same-sex friendships and a growing desire for independence and privacy.

5.7.2. Developmental milestones during Middle Adolescence (14-17 years) Ingimbi/ Umwangavu



This period involves continued physical development and body maturity with changes like acne and voice changes. Thinking becomes more advanced but may still be impulsive (without thinking carefully), while emotions are strong with identity confusion. Peer pressure is high, interest in relationships increases, and conflicts with parents and risk-taking may occur.

5.7.3. Developmental milestones during late adolescence (18-21)



Late adolescence is when physical growth is complete, and full maturity is reached. Thinking becomes mature and future-oriented, with better decision-making and emotional control. Individuals become more independent, form stable relationships, and experience less peer pressure.

Application activity 5.7



1. Who is the father of adolescent Psychology?
2. Describe the social development of early adolescents

5.8. Developmental milestones during adulthood (22-60years)

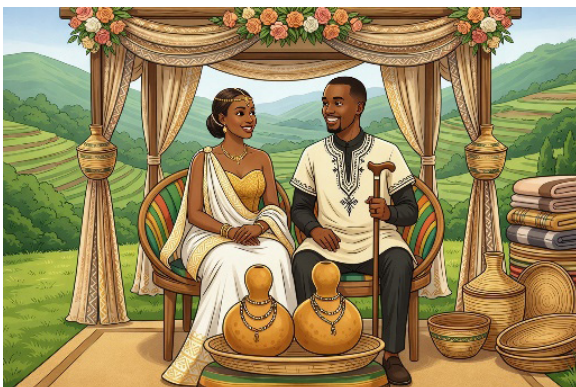
Learning activity 5.8



1. What is an adult?
2. What are the 3 categories of adulthood?
3. In Rwandan society, which things are not allowed to adolescents and children?

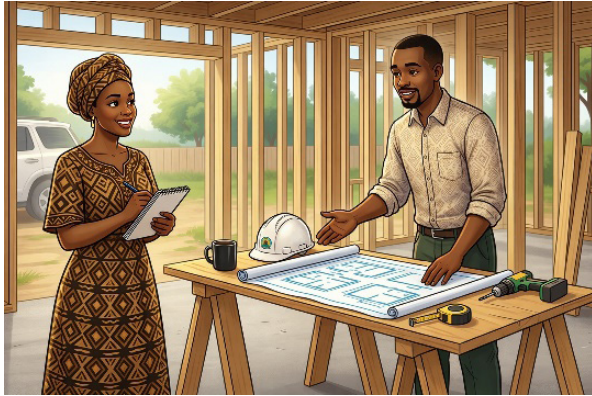
Developmental milestones during adulthood are the important changes that happen as a person grows older. According to Malcolm Knowles, a person becomes an adult biologically, legally, socially, and psychologically when they can take responsibility for their own life. Adulthood is divided into early adulthood (22–40), middle adulthood (40–60), and late adulthood (60+).

5.8.1. Developmental milestones during early adulthood (22-40years)



During this period an early adults become strong, think clearly, and become independent. They start careers, build close relationships, get married, and may have children. It is a time of growth in health, thinking, work, and family life.

5.8.2. Developmental milestones during middle adulthood (40-65)



This is a time of physical, mental, and social changes. The body slows down, with weaker muscles and bones, lower vision and hearing, and slower organ function.

Thinking slows too, but experience and practical problem-solving improve. Adults take on more responsibilities, care for family, contribute to society, and gain life satisfaction. Communication skills also improve, helping them express ideas clearly and maintain relationships.

5.8.3. Developmental milestones during late adulthood/old age (65 +)



This stage brings physical, mental, and social changes. The body becomes weaker, bones fragile, and vision and hearing decline. Thinking slows, but knowledge, wisdom, and life experience remain strong.

Many focus on family, friendships, and leaving a legacy, while losing people close to them and learn how to handle the sadness and changes that come with become common.

Communication stays important, often needing patience and clarity to stay connected with others.

Application activity 5.8



1. What are the physical changes during early adulthood?
2. Why people over 75 years old have significant hearing impairment.
3. Explain the key social-emotional milestone of middle adulthood.

5.9. End unit assessment



Section A: Knowledge and understanding

1. Define the following concepts:
 - a) Developmental milestone
 - b) Conception
 - c) Prenatal
 - d) Adolescence
2. Explain the difference between developmental milestones and developmental domains
3. What are the most common factors that women and infants are at risk during pregnancy and delivery in your community?
4. Explain how middle adulthood (40–60 years) differs from early adulthood physically, cognitively, and socially and emotionally.

Section B: Skills

5. Prepare a short speech explaining the physical, emotional, and social changes that happen during adolescence. Include advice on how to manage these changes and why it is important to handle them responsibly.

Section C: Attitudes

6. Using your knowledge of developmental milestones, explain why it is important to support both children and older adults. Describe ways you can help them in your community
7. Using an example, explain how parents, caregivers, and families should support a child's development before and after birth.

UNIT 6

FACTORS OF HUMAN DEVELOPMENT



Key Unit Competence:

Examine the factors of human development

Learning objectives

By the end of this unit, you should be able to:

Knowledge and understanding

- Define key terms related to human development such as nature, nurture, heredity, genes, and DNA.
- Identify the main factors influencing human development
- Describe how human development is influenced by heredity and environment
- Explain how heredity and environment work together to shape a human being
- Explain the negative impact of drug abuse on physical, psychological, and social development

Skills

- Compare and contrast biological and environmental factors which influence human development
- Interpret real-life cases or scenarios showing the interaction between biological and environmental factors.
- Debate on the effect of nature and nurture on human development
- Suggest possible solutions to improve human development conditions

Attitudes and values

- Demonstrates appreciation for diversity in human development
- Respect human beings regardless of their differences
- Value the role of a responsive community in supporting human development
- Develop a commitment to avoiding drug use and promoting drug-free lifestyles.
- Appreciate the role of family and society in supporting holistic human development

INTRODUCTION

Have you ever noticed any similarity in different people, but very different in others? Why do some skills, talents, or traits seem to run in families, while experiences and environment also shape who we are? Human development is influenced by many factors, including biological (heredity/nature) and environmental (nurture/experiences) influences.

In this unit, you will learn key concepts such as nature, nurture, heredity, genes, DNA, and environment, and explore how these factors work together to shape a human being. You will examine real-life examples of how biological factors influence characteristics like physical appearance, abilities, or personality, and how environmental factors, such as family, culture, school, and community, shape behaviour, social skills, and learning.

You will also study the negative impact of drug abuse on physical, psychological, and social development, and discover strategies for promoting healthy, drug-free lifestyles. Through activities like analysing case studies, creating comparison charts, and debating nature versus nurture, you will develop skills to interpret how these factors interact in real-life situations.

By the end of this unit, you will be able to explain the main factors that influence human development, appreciate diversity among individuals, and value the role of family and community in supporting holistic growth. This understanding will help you make informed decisions, respect differences, and promote a healthy, supportive environment for yourself and others.

Introductory activity



1. Using the proverbs 'Inyana ni iya mweru' and 'Mwene Samusure avukana isunzu' and the picture, give examples of ways a child can both look like and learn behaviours or skills from their parents.
2. What skills are being learnt by those children from their father?
3. Why do you think the father chose to make a carpentry workshop, not a car garage?
4. If these children moved to a big city where there are no trees or carpentry workshops, would they still develop the same skills? Explain.

6.1. Key terms related to factors of human development (nature, heredity, environment, nurture, genes, DNA)

Learning activity 6.1



1. What characteristics should an individual have from the parents or caregiver?
2. Why do you think the term “environment” is among the key terms related to factors of human development?
3. A judge needs ‘scientific proof’ that a suspect was at a crime scene because a single drop of blood was found on a broken window. Why does the police send this blood to the Rwanda Forensic Institute?

The following are the important words and ideas you need to understand this topic:

Nature refers to what you are born with, like your body and natural abilities from your parents. Nature looks at the impact of such physical approaches as neurotransmitters and genome sequencing on a child’s development

Nurture refers to what you learn and experience from your environment. These include things such as family care, education, culture, nutrition, friends, and the community. They shape a person’s behaviour, personality, and skills through learning and experience.

Heredity refers to how parents pass characteristics to their children through genes or the transmission of traits from one generation to the other. It influences all aspects of physical appearance such as height, weight, body structure, and the colour of eyes, the texture of the hair, and even intelligence and aptitudes. Diseases and conditions such as heart disease, diabetes, and obesity...can also be passed through genes, thereby affecting the growth and development of the child adversely

Chromosomes are structures in the cell that carry DNA and genes. Humans have 23 pairs of chromosomes that contain all the genetic information we inherit from our parents.

Genes refer to small units in your body that carry information from your parents and decide your characteristics.

DNA refers to the material in your body that contains genes. Deoxyribonucleic acid (DNA) is made up of genes and gives instructions for how your body grows and works.

Epigenetics refers to chemical changes that control how genes on the chromosomes work, without changing the DNA itself. For example, epigenetic changes may turn the height gene on or off, affecting how strongly the height gene is expressed.

Environment refers to everything around you (family, school, friends, and the place you live) that affects your development.

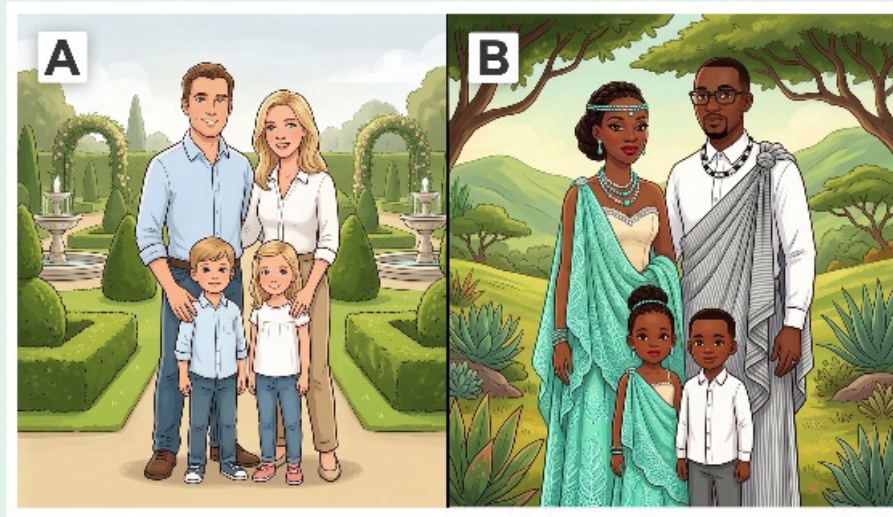
Application activity 6.1



1. What is DNA in full?
2. In which way do parents pass characteristics to their children?
3. How do we call the units that carry information from parents?
4. Differentiate nature from nurture.
5. What do you understand by the term "environment" in human development?

6.2. Biological factors (nature/heredity)

Learning activity 6.2



Observe the picture above and answer the following questions,

1. What is the difference between Family A and Family B?
2. Why do the children have different skin colours and hair?
3. How do you think a child's physical appearance and genetic traits get influenced when a black parent marries a white parent?

All species transmit characteristics from one generation to the other through the mechanisms of genetics. It implies that every human being carries genetic traits which are inherited from parents. Those genetic traits are located within every cell in our body. Heredity is therefore the transmission of traits from one generation to the other. Heredity plays a key role in human development through genes passed from parents and grandparents.

When sperm and ovum combine, they determine physical traits like the colour of eyes, colour of hair, straight or curly hair, pigmentation of the skin, and certain diseases. Many physical characteristics are strongly influenced by genes inherited from biological parents, even if they are not completely determined. These include traits like height,

weight, hair loss in men, life expectancy, and vulnerability to certain illnesses such as breast cancer. This has led people to wonder whether psychological traits like behaviour, personality, and intelligence are also influenced by heredity before birth.

Function of genes and chromosomes in determining traits

Genes are parts of DNA that carry instructions for traits, like eye colour, height, or the chance of getting certain diseases. These genes are arranged on structures called chromosomes. Humans have 46 chromosomes (23 pairs), with one set from each parent. Traits are passed from parents to children. Girls get an X chromosome from each parent, while boys get an X from their mother and a Y from their father. Some genes are dominant, which means their trait shows even if only one copy is present, while recessive genes show their trait only if both copies are the same. Traits can be influenced by many genes and by the environment, and recessive traits can sometimes skip generations.

Application activity 6.2



1. Outline 3 examples of physical characteristics which are determined by biological factors
2. How does a baby inherit chromosomes from her/his parents?
3. Explain the relationship between genes and nature/heredity
4. How does nature influence human development?

6.3. Environmental factors (nurture/experiential)

Learning activity 6.3



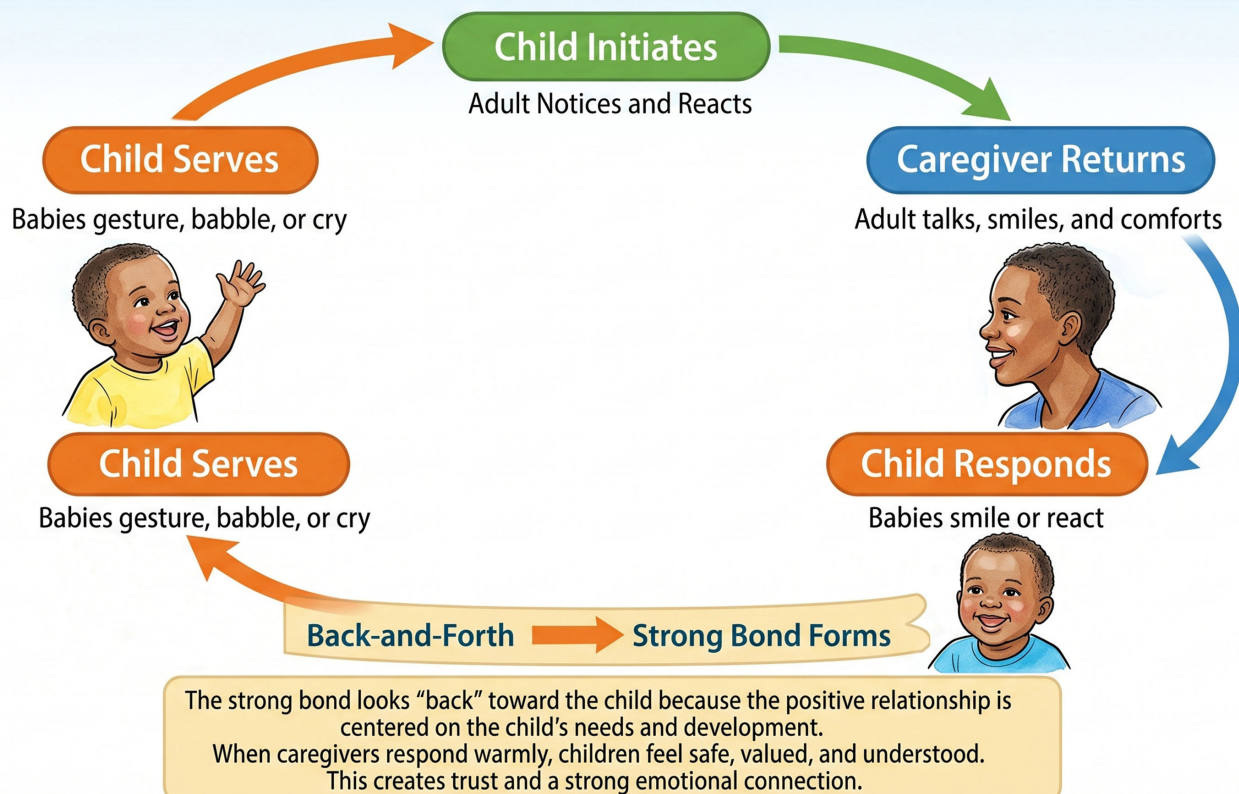
Nyirarukundo is a working mother who has a 9-month-old baby. In her absence, the baby stays with Mutesi, the caregiver. Mutesi is a good caregiver who is very responsive to the baby's needs... The baby always enjoys staying with Mutesi. The baby would cry in her (Mutesi) absence instead of her mother. She (the baby) even cries when in his/ her mother's hands.

1. Why does the baby cry when in her mothers' hand?
2. What advice would you give to other mothers to avoid the same situation?

Human development is sharpened by things a person lives through or is exposed to. These external factors start during the prenatal period, influencing gene expression via epigenetics, and they include all non-genetic influences shaping human development.

Impact of relationships (serve and return)

Serve and Return Interaction



Source: Centre on the Developing Child at Harvard University. *Serve and Return Interaction in Early Childhood Development*.

Researchers explain that this interaction works like a game of tennis, where the child "serves" and the adult "returns." These exchanges help build brain connections and support language, social skills, and emotional development.

Brain research shows that early experiences affect how the brain develops. And the most important factor for a child is to have loving relationships with a caregiver. Children learn about the world through serve and return relationships with parents, family, and caregivers, which help them develop thinking, communication, emotions, behaviour, and social skills. They also learn by observing how their parents interact with others, gaining models of kindness, respect, and social behaviour. By responding in a warm, loving, and gentle way, they help their child learn communication, emotions, and behaviour, while making them feel safe and secure. Feeling safe provides children with the confidence to explore the world, develop resilience, and prepare for challenges, healthy relationships, and better mental health later in life.

Application activity 6.3



1. What is meant by "serve and return" relationships between a child and a caregiver
2. Explain three examples of what caregivers or parents should do and three examples of what they should not do to support a child's healthy development.

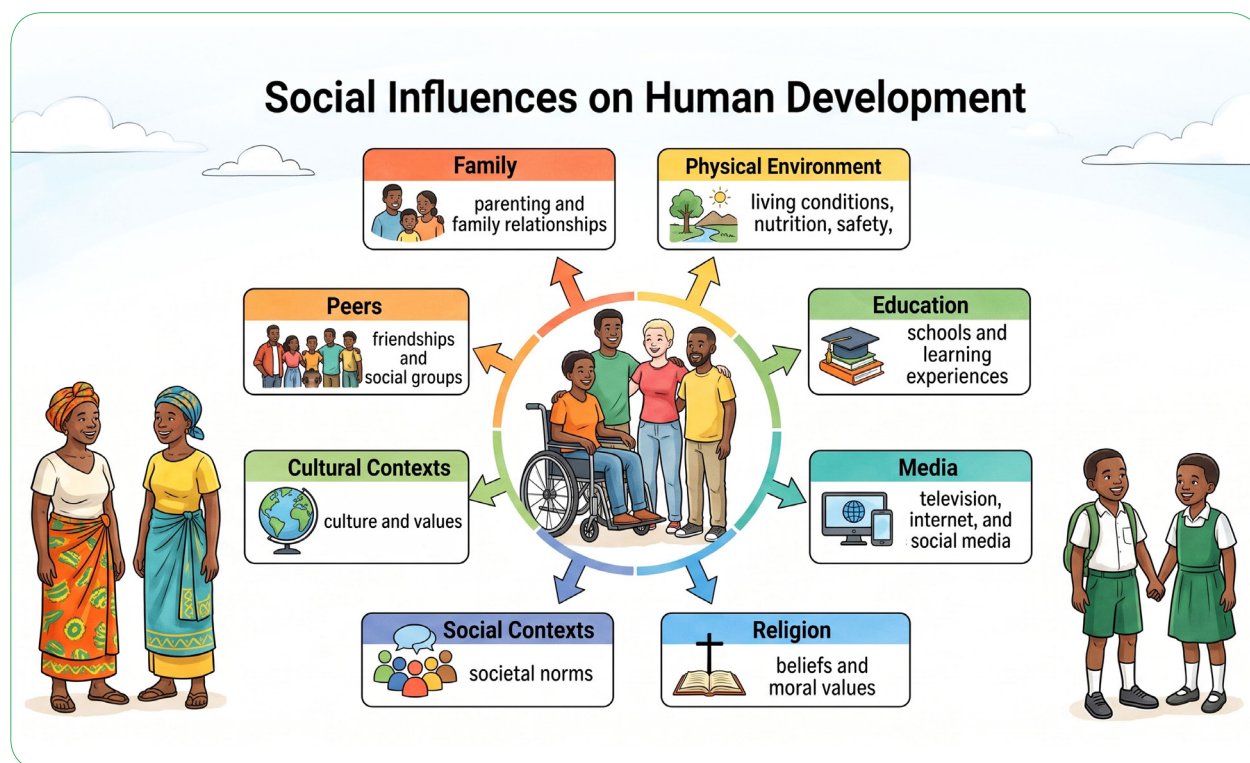
6.4. Influence of socialising agents on human development

Learning activity 6.4



What do you understand by the proverb that says: “Mbwira uwo mugendana, ndakubwira uwo uri we (in English: ‘Tell me who your friends are, and I’ll tell you who you are’/ ‘A man is known by the company he keeps’ or in French: Dis-moi qui te hante, je te dis celui qui tu es).

Socialising agents on human development are people, groups, or institutions that influence how individuals learn behaviours, values, attitudes, and social skills as they grow and interact with society. They help shape a person’s personality, beliefs, and way of behaving by teaching social rules, cultural values, and appropriate ways of interacting with others. These socialising agents include: family, school, peers, media, beliefs, and the community, which all contribute to a person’s social, emotional, and cognitive development.



Source: Nickerson, C. (2026). Agents of Socialisation. Simply Psychology.

a) Family environment

Parenting styles, family relationships, socioeconomic status, and early life stress all play important roles in child development. Supportive and warm parenting helps children build confidence, social skills, and emotional health, while harsh or neglectful parenting can cause behavioural or emotional problems. Positive family relationships provide security, role models, and guidance, teaching children how to interact respectfully and solve problems. Socioeconomic status also affects access to resources like food, healthcare, and education, which influence physical and cognitive growth.

Early life stress, such as trauma or instability, can impact brain development and emotional regulation. Together, these factors shape a child's ability to learn, adapt, and thrive in life.

b) Physical environment

The physical environment plays a key role in a child's health and growth. Proper nutrition provides the essential energy and nutrients needed for strong bones, muscles, and overall development. Exposure to environmental toxins, such as pollution or harmful chemicals, can damage the body and slow growth. Sunlight is important for vitamin D production, which supports bone health and immunity. Safe and stable housing ensures children are protected from hazards and illness, allowing them to grow and thrive.

c) Social and cultural contexts

Human behaviour and personality are shaped by the social and cultural environment around them. From friends and peers, they learn how to communicate, cooperate, and handle challenges, while schools provide structure, knowledge, and guidance that build confidence, discipline, and problem-solving skills. At the same time, societal values teach children what is considered acceptable and ethical behaviour, and cultural norms influence how they express themselves, relate to others, and understand their roles in the family and community. Together, these influences create a strong foundation for children to develop attitudes, social skills, and personality traits that guide their behaviour throughout life.

d) Media

Mass media like TV, newspapers, radio, and the Internet influence children's behaviour, personality, values, and emotions. Along with economic, political, and cultural institutions, it helps guide how children think, socialise, and live their lives.

e) Belief

Organised religion or belief teaches people important social values and helps them learn how to behave in society. Religious ceremonies, like weddings and births, support family traditions, and some practices show expected roles for men and women. Through these teachings and events, religion helps pass shared values from one generation to the next.

f) Education opportunities

Schools do more than teach subjects like math and science; they also shape children's behaviour and social skills. Through lessons, routines, classroom rituals, and teacher role models, schools teach teamwork, discipline, respect, and problem-solving. Interacting with peers and participating in activities helps children build confidence, social skills, and mental wellbeing.

g) Peer

Peer groups are people of similar age and social status who share interests. Children start learning social rules from peers early in life, such as taking turns or following game rules. As they grow into teenagers and adolescents, peer groups become more important because they help them develop independence and form their own identity.

Peer groups also provide social experiences outside the family. However, even though friendships are very important to adolescents and teenagers, parents still continue to influence them.

h) The Workplace

Just as children spend much of their day at school, adults spend a lot of time at work. Even though they are already familiar with their culture, workers need to learn the rules and ways of behaving in a workplace. This includes practical skills, like using equipment, and social rules, like how to communicate with a boss. Different jobs require different types of socialisation.

Application activity 6.4



1. What is the first socialisation agent that the child meets?
2. Think about your life journey from your youngest age to now and write down on a piece of paper how nurture affected your development socially and cognitively.

6.5. Impact of drug abuse on human development

Learning activity 6.5



1. Observe the above picture and describe the behaviour you learn from it.
2. Imagine one of the parents in this family used to come home drunk and fighting; how does such behaviour affect the children's development?

Drugs are substances that can change how your body or mind works. They can be used for medicine (to treat illnesses) or can be abused (alcohol, tobacco, cocaine, urumogi, kanyanga, etc), which can affect your health, behaviour, and thinking. Chronic drug use can cause serious mental and physical health problems, such as depression, anxiety, hallucinations, high blood pressure, stroke, and heart disease. Many people with addiction also have other mental disorders.

Signs of drug addiction

Changes in appearance (like weight loss), changes in behavior with family or friends (becoming aggressive or withdrawn), neglecting work responsibilities (being late or absent), and reducing time spent on hobbies or social activities, feeling strong urges to use drugs, failing to stop, needing more to feel the same effect, and keeping drugs even if they cannot afford them are most common signs of drug addiction

Drug abuse effect

Drug abuse can slow or harm a person's development in many ways. It can damage the body and brain, causing problems like heart disease, high blood pressure, liver and kidney disease, and affecting energy. It can make thinking, learning, and decision-making harder. It can cause emotional problems like depression, anxiety, paranoia, or hallucinations. It can also hurt relationships and social skills, making it difficult to meet responsibilities and grow as a person.

Drug Addiction Treatment

Drug addiction treatment should be tailored to a person's individual needs to be effective. Treatment may include behavioural therapy to build coping skills, group therapy to work through psychological challenges with peers, and medications to reduce withdrawal symptoms. Additional support may include medical care, vocational training, and resources to address problems like mental health issues or unemployment. Many of these services are provided in rehabilitation centres, where individuals receive structured care and guidance for recovery.

Application activity 6.5



1. What is drug abuse?
2. What are the effects of drug abuse on human cognitive development?
3. Why does the Rwandan government take addicted people to drug rehabilitation centres instead of prisons?

6.6. End unit assessment



Section A: Knowledge and understanding

1. What do you understand by the following terms:
 - a) Nature
 - b) Nurture
 - c) Epigenetic
 - d) Genes
 - e) Heredity
2. Identify 4 biological traits that can be inherited from the parents

3. Explain what is meant by environment in human development.
4. Differentiate nature from nurture
5. Describe 2 ways drug abuse can affect human development.

Section B: Skills

6. Compare biological factors and environmental factors that influence human development.
7. *"Uwera grows up in a family where parents encourage reading, respect others, and attend school regularly".* Analyse how Uwera's family environment may influence her behaviour and personality.
8. Suggest 3 ways teenagers can resist negative peer pressure.
9. In which way can you help a young man recover from drug abuse and improve his development?

Section C: Attitudes and values

10. How does your understanding of nature, nurture, heredity, and environment helps appreciate that people develop differently and respect others regardless of their differences?
11. Describe how a young person can use social media responsibly to support positive growth and healthy development.
12. Explain how a supportive environment can help young people develop responsibly.

7. PROJECT ACTIVITY

❖ Project title:

The influence of school environment on adolescents' development

❖ Objectives: By the end of this project, you will be able to:

Knowledge	Skills	Attitudes and Values
Explain how the school environment influences adolescent development.	Apply psychological knowledge to a real-life situation.	Appreciate the importance of a supportive school environment for adolescent development.
Identify factors that influence human development.	Apply relevant schools and branches of Psychology to explain real-life situations.	Demonstrate respect for adolescents' individual differences and developmental needs.

Knowledge	Skills	Attitudes and Values
Explain adolescent behaviour and development using relevant theories of human development.	Analyse factors influencing adolescent development.	Value positive relationships and interactions within the school community.
	Propose practical ways of creating a supportive school environment.	Demonstrate responsibility in contributing to a safe and supportive school environment.

❖ Project activity:

As an S4 student, investigate how the school environment can influence the development of adolescents. Identify factors in the school environment that may positively or negatively influence students' physical, cognitive, emotional or social development. Using the psychological knowledge you have acquired throughout the year, explain these influences and suggest practical ways of creating a school environment that supports healthy adolescent development.

❖ Project outline

After conducting your research and analysing the information collected, write a report presenting your findings and psychological interpretation. Your report should include the following sections:

- 1. Introduction:** Briefly introduce the topic and explain the purpose of the research.
- 2. Methodology:** Explain how you collected the data, including the methods used, the respondents and the research setting.
- 3. Findings:** Present the main findings from the data collected. Use tables, graphs, percentages or brief explanations where appropriate.
- 4. Psychological Analysis:** Interpret your findings using the psychological knowledge you studied, including relevant schools or branches of Psychology, theories of human development, and factors influencing development.
- 5. Conclusion:** Summarise the main findings and what they show about adolescent development.
- 6. Recommendations:** Suggest practical ways to address the challenges identified and promote healthy adolescent development.
- 7. Reflection:** Briefly explain what you learned from conducting the research and how the project helped you apply Psychology to a real-life situation.

❖ Guiding questions:

Your research should consider the following questions:

2. Which factors in the school environment appear to have the greatest influence on students' development?
3. How do these factors influence students' development? Give examples based on your findings.

4. Are the effects of these factors positive, negative, or both? Explain your answer.
5. Which area(s) of human development; physical, cognitive, emotional, or social, are most affected by these factors? How?
6. Which theory of human development studied in this book can help explain your findings? How does the theory apply to your findings?
7. Which school or branch of Psychology studied in this book can help explain the situation? How is it relevant to your findings?
8. What challenges related to the school environment/have you identified from your findings?
9. What practical measures can be taken to address the identified challenges and promote healthy adolescent development?
10. What conclusion can you draw about the influence of the school environment on adolescent development?
11. Based on your findings, what recommendations would you give to the policy makers, school leadership, students, parents/caregivers, etc.

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